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FEATURE ARTICLE

Theoretical studies of reactions of carbon dioxide mediated and catalysed by transition metal complexes

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In this Feature article, we have reviewed theoretical studies of transition-metal mediated and catalysed reactions involving carbon dioxide. We have discussed fundamental steps related to organometallic reactions of carbon dioxide and presented and analyzed reported theoretical studies of transition-metal mediated and catalysed reactions of carbon dioxide.

1. Introduction

Carbon dioxide has been recognized as the major contributor to the greenhouse effect, causing global warming in the atmosphere.[1,2] The emission level of the greenhouse gas keeps increasing because of the excessive usage of fossil fuel in the environment. An effective chemical solution is to convert it into new chemical products. As an attractive C1 building block, carbon dioxide is abundant, cheap, nontoxic, and nonflammable.[3] However, chemical transformation of carbon dioxide presents significant challenge for chemistry due to the fact that it is the most oxidized of carbon, thermodynamically stable and kinetically inert. Despite its thermodynamic stability, thermodynamically favorable processes or reactions can still be achieved when reactive reagents are employed. Over the past years, progress has been made in exploring chemical reactions of carbon dioxide. A large number of its reactions with transition metal complexes have been reported.[4,5] In many of these reported reactions, carbon dioxide can be transformed into other organic products when proper transition metal complexes are employed.

Along with experimental studies, there have also been a number of theoretical studies on reactions of carbon dioxide mediated or catalysed by transition metal complexes. In this feature article, we give an up-to-date discussion of the current theoretical understanding of various reactions of carbon dioxide with transition metal complexes. We will first discuss some fundamental steps related to organometallic reactions of carbon dioxide, then present reported theoretical studies of transition-metal mediated reactions, and finally focus on transition-catalysed reactions. We do not intend to provide a compressive review. Instead, we selected topics that we believe to be important to the activation of carbon dioxide from our own perspective.

Coordination of CO_2 to a metal center is normally a necessary event for its conversion to other chemicals mediated or catalysed by transition metal complexes. Three most popular coordination modes are shown in Scheme 1. Theoretical studies on the relative stability of these coordination modes have been widely reported.[6] Up to date, several review

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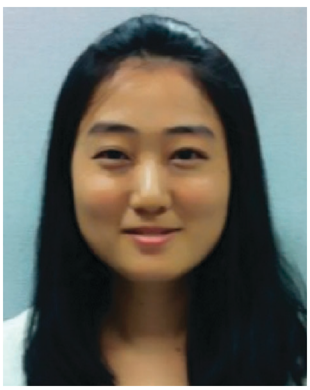

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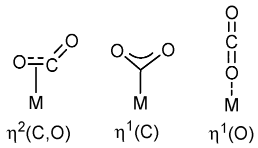

Scheme 1

articles can also be found in the literature dealing with the coordination chemistry issue.[3a,5e,f] Here, we refer interested readers to these review articles for the coordination chemistry of CO_2. The current work mainly deals with the reaction aspects.

2. Theoretical framework

This Feature article concerns theoretical investigations of reactions involving carbon dioxide. Thus, results (pertaining to energetics) from various levels of theoretical calculations will be discussed. Given the large amount of theoretical work reviewed here, variations in theoretical methodology are expected. It is known that different levels of theoretical calculations could lead to significantly different results,[7] making it difficult to compare. Cases of this type are often seen when we deal with very weak interactions such as van der Waals forces because dispersion forces are not properly taken care of in popularly employed DFT methods.[8] Here, we discuss processes related to bond breaking and formation in carbon dioxide, which involve very strong interaction forces. Furthermore, theoretical calculations reviewed in this article, except those reported in the early 1990s, are almost uniformly performed within density functional theory framework. Thus, we can make a reasonable assumption that the energetic data calculated and reported in the literature for reactions of CO_2 with transition metal complexes are comparable.

A second issue that deserves brief comment is the accuracy of theoretical calculations. In this article, activation energies that are theoretically calculated will be constantly discussed and compared. We know that the accuracy of theoretical calculations can only be best assessed from direct comparison between theoretical predictions and experimental results in well-defined systems. There are some experimental measurements of kinetic and thermodynamic parameters for systems involving reactions of CO_2 with transition metal complexes.[9,10] However, reports on direct comparison between theoretical and experimental results are very limited in the literature. One recent example of this type was reported by Lee, Rybak-Akimova, Holm and their co-workers on kinetics and mechanistic analysis of an extremely rapid carbon dioxide fixation reaction by 2,6-pyridinedicarboxamidatonickel(II)–hydroxide complexes.[11] In this work, activation enthalpy and entropy of 3.2(5) kcal mol^{-1} and −20(3) cal mol^{-1} K^{-1}, respectively, were experimentally obtained. Theoretical calculations at various levels of density functional theory (B3LYP and BP86 with different basis sets) were also performed. The calculated activation enthalpies ($\Delta H^{\ddagger}$) are comparable with the experimentally obtained value, with the BP86 and B3LYP results slightly higher and lower by 1–2 kcal mol^{-1}, respectively, than experiment. However, the calculated activation entropies ($\Delta S^{\ddagger}$) are consistently more negative than experiment by *ca.* 14–17 cal mol^{-1} K^{-1}. We can see here that the activation entropy contribution ($-T\Delta S$) to the activation free energy was experimentally estimated to be around 6 kcal mol^{-1} at 298.15 K, but theoretically to be 10–11 kcal mol^{-1}. In other words, if activation free energies are used in discussion, a theoretical overestimation of 4–5 kcal mol^{-1} from the entropy contribution alone is likely to be expected when we deal with two-to-one (bimolecular) transformations. For one-to-one transformations, the values of $\Delta G^{\ddagger}$, $\Delta E^{\ddagger}$ and $\Delta H^{\ddagger}$ are approximately the same because of small entropy changes.

A third issue is whether calculated activation free energies or activation enthalpies should be used to discuss energy profiles of reactions. Since the reactions we will deal with involve the gaseous molecule CO_2, we believe that using calculated free energies to discuss the reactions is more appropriate. However, a small amount of theoretical work reported in the literature does not provide data related to free energies. In such cases, we estimate $\Delta G^{\ddagger}$ values by approximately adding 10 kcal mol^{-1} to calculated $\Delta E^{\ddagger}$ (or $\Delta H^{\ddagger}$) for two-to-one (bimolecular) transformations in consideration of entropy contribution. Addition of 10 kcal mol^{-1} is based on the above-discussed results by Lee, Rybak-Akimova, Holm and their co-workers[11] and our own extensive experiences. Here, one may question why we do not add 6 kcal mol^{-1}, a value estimated experimentally. The answer to this question is that throughout the article we focus our discussion on reported theoretical results. Addition of 10 kcal mol^{-1} would make the results comparable. An implication for readers is that they should pay attention to the relative heights of activation free energy barriers discussed, not to the absolute activation free energy barriers presented. Computational errors exist in those activation free energy barriers calculated and reported. However, the relative values should be reliable because computational errors are cancelled when relative barrier heights are considered.

Zhenyang Lin

Zhenyang Lin received his BSc in Chemistry from China University of Geosciences in 1982 and his MPhil degree in 1985 from the Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, where he worked with Professors Jiaxi Lu and Chunwan Liu. His doctoral research was carried out from October 1986 to December 1989 in the Inorganic Chemistry Laboratory of the University of Oxford under the supervision of Professor D. Michael P. Mingos. After his postdoctoral work with Professor Michael B. Hall at Texas A&M University, he joined the faculty at the Hong Kong University of Science and Technology in 1994. He is now the Head of the Chemistry Department. His research interests include theoretical aspects of structure, bonding and reactivity of inorganic and organometallic compounds, and homogeneous catalysis.

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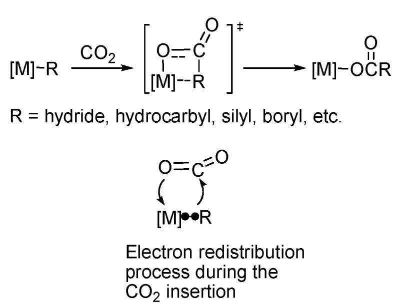
3. CO_2 insertion into an M–R bond

A number of theoretical studies have been reported on the mechanism of CO_2 insertion into an M–R bond (R = hydride, hydrocarbyl, silyl, boryl, *etc.*). The results of these theoretical studies indicated that the insertions proceed *via* a variety of transition state structures. Majority of the insertion mechanisms consider the R ligand as the nucleophile initiating nucleophilic attack at the carbon center of CO_2.

3.1 CO_2 insertion into a metal–hydrocarbyl bond

As early as in the 1990s, Sakaki and coworkers investigated the CO_2 insertion into the $(PH_3)_2$Cu–R (R = CH_3 or H) bond with theoretical calculations using the single and double configuration interaction (SD-CI) method.[12] They discussed the nature of the insertion reaction by emphasizing the nucleophilic attack of the Cu–R σ bond at the carbon center of CO_2 during the insertion process. The transition state structures show a four-membered ring structure illustrated in Scheme 2. The activation energy ($\Delta E^{\ddagger}$ = 4.2 kcal mol^{-1}, $\Delta G^{\ddagger}$ ≈ 14.2 kcal mol^{-1}) of the CO_2 insertion into the Cu–H bond was found to be lower than that ($\Delta E^{\ddagger}$ = 9.3 kcal mol^{-1}, $\Delta G^{\ddagger}$ ≈ 19.3 kcal mol^{-1}) of the CO_2 insertion into the Cu–CH_3 bond. The lower energy barrier for insertion into Cu–H is because the spherical 1s valence orbital of H, when compared with the directional sp^3 hybridized valence orbital of CH_3, makes $(PH_3)_2$Cu–H distort less than $(PH_3)_2$Cu–CH_3 to get to the transition state structure.

Very recently in our theoretical studies on the carboxylation of arylboronate esters with CO_2 catalysed by copper(I) complexes, we systematically examined the CO_2 insertions into Cu–C(phenyl, vinyl, ethyl, ethynyl) bonds.[13] The CO_2 insertion into the Cu–C(vinyl) bond *via* the normal insertion mode shown in Scheme 2 was found to have approximately the same free energy barrier (*ca.* 20 kcal mol^{-1}) as that into the Cu–C(phenyl) bond. The CO_2 insertion into the Cu–C(ethyl) bond was calculated to be kinetically less favorable, with a free energy barrier of 24.1 kcal mol^{-1}. Interestingly, the CO_2 insertion into the Cu–C(ethynyl) bond has the largest free energy barrier (30.0 kcal mol^{-1}). The CO_2 insertions into Cu–C(phenyl, vinyl, ethyl) bonds are slightly exergonic by 2–6 kcal mol^{-1} while the insertion into the Cu–C(ethynyl) bond is endergonic by 15 kcal mol^{-1}. Molecular orbital analysis of the transition state for the insertion processes shows that the Cu–C σ bond nucleophilically attacks the electron-deficient CO_2 carbon. In the CO_2 insertions into Cu–C(phenyl, vinyl, ethynyl) bonds, the π bonding molecular orbitals of the phenyl, vinyl and ethynyl ligands also contribute to the stability of the transition state by interacting with the metal center. The calculation results related to the barriers reported above were explained as follows. Higher barriers for the insertions into Cu–C(sp^3) and Cu–C(sp) bonds were, respectively, related to the lack of a π bonding molecular orbital in the alkyl ligand and to the very strong Cu–C σ bond for the alkynyl ligand.



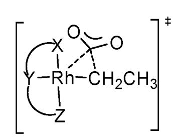
Scheme 2

Regarding insertion into an M–C(sp^3) bond, another interesting insertion mode has been found on the basis of theoretical calculations. Schmeier, Hazari and coworkers found in their theoretical calculations that CO_2 insertion into the (PCP)Ni–CH_3 bond (PCP = 2,6-$C_6H_3(CH_2PMe_2)_2$) occurs *via* an S_E2-like mechanism through CO_2 electrophilically attacking the methyl ligand from the open side and then undergoing rearrangement to give the insertion product.[14] This insertion mode results in an inversion of the C(sp^3) stereocenter. The activation free energy *via* such an unusual mode was calculated to be 30.2 kcal mol^{-1}. In 2012, the same group calculated CO_2 insertion into the (PCP)Ni–C≡CPh bond (PCP = 2,6-$C_6H_3(CH_2PMe_2)_2$) and found that CO_2 directly attacks the α carbon of the acetylide ligand *via* an S_E2-like mechanism.[15] The activation free energy was calculated to be 30.6 kcal mol^{-1}.

More recently, Hölscher, Leitner and coworkers studied theoretically the ligand effect on CO_2 insertion step using the 16-electron square planar model complexes LRh-CH_2CH_3.[16] Here L is a tridentate pincer ligand coordinating to the metal center in a meridinal mode. The transition state shows a three-membered ring structure shown in Scheme 3. It was found that the central donor atom of the tridentate pincer ligand *trans* to ethyl ligand has a greater influence on the insertion barrier than the donor atoms *cis* to it. It was also found that anionic complexes were more reactive than the neutral ones. Molecular orbital analyses of the model complexes allowed the authors to make the following observation. Activation of CO_2 occurs *via* the orbital interaction of the HOMO of a given model metal complex and one of the LUMOs of CO_2. If the HOMO consists of the orbital interaction between Rh d_{z^2} and the Rh-bonded C p_z of the ethyl ligand, the activation free energy ($\Delta G^{\ddagger}$) is found to be within 25 kcal mol^{-1}. Here, the *z*-axis is defined as being perpendicular to the square plane of the model complex. If the HOMO consists of the orbital interaction between Rh d_{xy} and the Rh-bonded C p_z of ethyl, the activation free energy will be greater than 25 kcal mol^{-1}.

3.2 CO_2 insertion into a metal–allyl bond

In 1997, Shi and Nicholas reported an experimental study of Pd-catalysed CO_2 insertion into Sn–allyl bonds to form tin carboxylates which are widely used in industry as stabilizers.[17] Interestingly, they found that tetrabutyltin, tetraphenyltin, vinyltribytyltin and benzyltributyltin did not react. These results



Scheme 3

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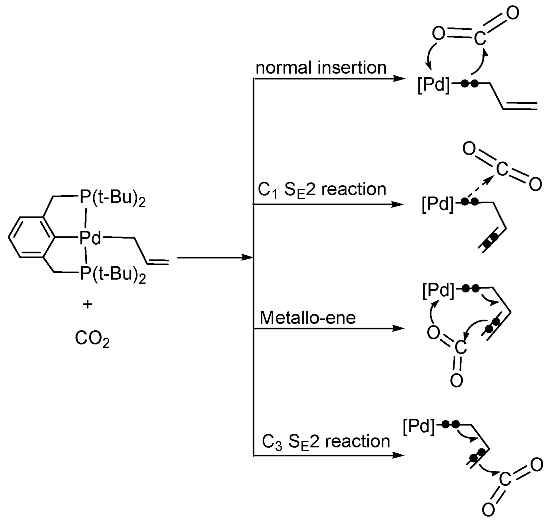
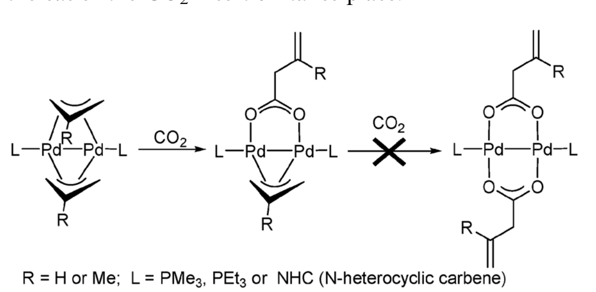


Scheme 4

indicated that the reactivity is closely related to the property of Pd–allyl bonds. There are several theoretical studies on the mechanism of CO_2 insertion into Pd–allyl bonds, which will be discussed below.[18]

Wendt and coworkers carried out a series of mechanistic probe experiments to find how CO_2 inserts into a Pd–η^1-allyl bond upon reacting square planar complexes (PCP)Pd-(η^1-allyl) (PCP = 2,6-bis[(di-*tert*-butylphosphino)methyl]-phenyl) with CO_2.[18a] They concluded that a normal insertion mechanism shown in Scheme 2 is impossible. They also carried out DFT calculations on four different modes of CO_2 insertion into the Pd–η^1-allyl bond using a simplified model complex shown in Scheme 4. The normal insertion mode was calculated to have an activation barrier of $\Delta H^\ddagger$ = 29.3 kcal mol^{-1} ($\Delta G^\ddagger \approx$ 39.3 kcal mol^{-1}). The C^1 S_E2 mode was calculated to have an activation barrier of $\Delta H^\ddagger$ = 21.7 kcal mol^{-1} ($\Delta G^\ddagger \approx$ 31.7 kcal mol^{-1}). The metallo-ene mode with a six-membered ring transition state considers the η^1-allyl π bond as the nucleophile attacking the CO_2 C atom and the activation barrier was calculated to be $\Delta H^\ddagger$ = 14.7 kcal mol^{-1} ($\Delta G^\ddagger \approx$ 24.7 kcal mol^{-1}). The C^3 S_E2 mode also considers the η^1-allyl bond as the nucleophile and was found to have the lowest activation barrier ($\Delta H^\ddagger$ = 14.2 kcal mol^{-1}, $\Delta G^\ddagger \approx$ 24.2 kcal mol^{-1}). The lowest activation energy was attributed to that the Pd–η^1-allyl complex requires least rearrangement to reach the transition state *via* this insertion mode.

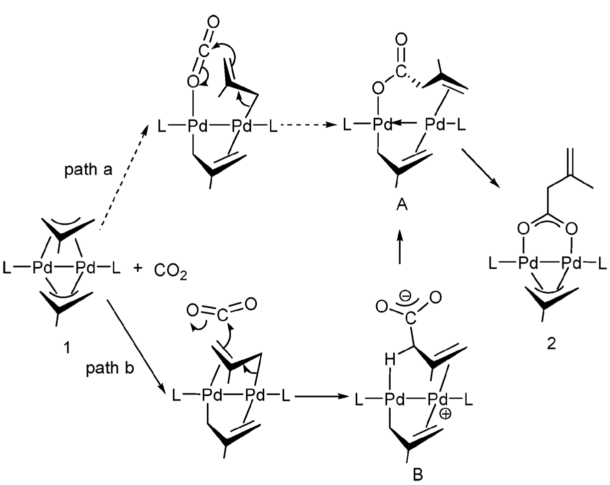
In 2010, Hazari, Schmeier and coworkers studied experimentally the reactions of CO_2 with (η^1-allyl)(η^3-allyl)PdL (L = PMe_3, PEt_3, PPh_3, NHC) and found that CO_2 only inserts into the Pd–η^1-allyl bond, not into Pd–η^3-allyl.[18b] They also carried out DFT calculations to study the effects of different ligands (PH_3, PMe_3, NHC) on the CO_2 insertion into Pd–η^1-methylallyl and Pd–η^1-allyl bonds. *Via* the most favorable C^3 S_E2 insertion mode, the activation barrier of CO_2 insertion into a Pd–η^1-methylallyl bond was calculated to be lower by *ca.* 1.0 kcal mol^{-1} than that of CO_2 into a Pd–η^1-allyl bond. Taking the CO_2 insertion into a Pd–η^1-allyl bond as an example, the activation free energy increases in the order of PMe_3 (8.6 kcal mol^{-1}) < NHC (9.6 kcal mol^{-1}) < PH_3 (12.4 kcal mol^{-1}). These calculation results are consistent with the notion that the more electron-rich the metal–allyl bond is, the easier the CO_2 insertion takes place.



(1)

Recently, we carried out DFT studies on the insertion reactions of CO_2 with allyl-bridged dinuclear palladium(I) complexes (eqn (1)).[18c] Experimentally, it was found that although the allyl-bridged dinuclear palladium(I) complexes have two bridging allyl ligands, only one bridging allyl ligand is transformed into a bridging carboxylate ligand (eqn (1)).[19] With PMe_3 as the ligand L and 2-methylallyl as the bridging allyl ligand in our calculations, our DFT studies on the two possible pathways shown in Scheme 5 indicated that path b is the favorable pathway for the insertion reaction. Path a was calculated to have an overall free energy barrier of 42.9 kcal mol^{-1} while path b has an overall free energy barrier of 28.4 kcal mol^{-1}. In path b, CO_2 approaches one of two bridging allyl ligands in a direction perpendicular to the allyl plane at one of the allyl terminal. In other words, CO_2 is activated by the π electrons of the allyl ligand. Through our DFT studies, we were also able to explain why the second bridging allyl ligand does not further react with CO_2. The two allyl bridging ligands in the dinuclear palladium complexes have strong *trans* influence, which weakens the metal–allyl bond to facilitate the first CO_2 insertion. When one of the bridging allyl ligands was transformed into a bridging carboxylate ligand, the remaining bridging allyl ligand showed remarkably strong metal–allyl bonding interactions, impeding the second CO_2 insertion.

It should be pointed out here that Hazari, Schmeier and coworkers also carried out theoretical calculations of the CO_2



Scheme 5

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insertion into the allyl-bridged dinuclear palladium(I) complexes.[20] Overall, the calculation results obtained are similar to what we have described above. However, they found that CO_2 approaches one of the bridging allyl ligands in a direction lying in the allyl plane. We believe that this type of activation mode is very unlikely because CO_2 needs to be activated by π electrons of the allylic ligands.

3.3 CO_2 insertion into a metal–silyl, –germyl, or –stannyl bond

In 2008, Bhattacharyya, Sadighi and coworkers found experimentally that CO_2 reacted with (IPr)CuSnPh$_3$ (IPr = 1,3-bis(2,6-diisopropylphenyl)imidazole-2-ylidene) to form (IPr)CuOCOPh and SnPh$_2$, instead of the expected insertion product (IPr)CuOCOSnPh$_3$.[21] This experimental work prompted Ariafard, Yates and coworkers in 2010 to systematically study the CO_2 insertion into the Cu–E bonds of (NHC)CuEPh$_3$ (E = Si, Ge, Sn) theoretically.[22] To understand the experimental finding, they also examined activation of CO_2 by considering nucleophilic attack of a Ph group of EPh$_3$. In their calculations, the model complexes (NHC)CuEPh$_3$ (NHC = 1,3-dimethylimidazol-2-ylidine) were employed. When the nucleophilic attack of a Cu–E bond at the carbon center of CO_2 *via* the normal insertion shown in Scheme 2 was considered, the activation free energies were calculated to be 27.0, 31.3, 32.1 kcal mol^{-1} for E = Si, Ge, Sn, respectively. When the nucleophilic attack of a Ph group of EPh$_3$ at the carbon center of CO_2 *via* a concerted mechanism shown in Scheme 6 was considered, the activation free energies were calculated to be 47.2, 41.5, 32.9 kcal mol^{-1} for E = Si, Ge, Sn, respectively. On the basis of these calculated activation free energies, the following conclusion was made. The reactivity of the Cu–E bond decreases and the reactivity of the E–C bond increases as E becomes heavier.

3.4 CO_2 insertion into a metal–boryl bond

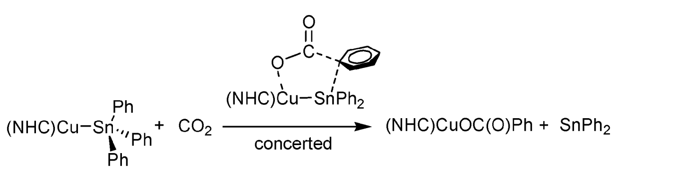
In 2006, we studied theoretically the mechanism of the reduction of CO_2 to CO catalysed by NHC copper(I) boryl complexes, which was reported by Sadighi and coworkers.[23] One of the important steps in the reaction mechanism is the CO_2 insertion into the Cu–B(boryl) bond. Although boryl ligands have formally an "empty" p orbital on the boron center, we found that the boryl ligands in the catalytic reactions are nucleophilic, not oxophilic. The CO_2 insertion into a Cu–B(boryl) bond occurs *via* a mechanism similar to what we have discussed in Scheme 2. When (NHC)CuBeg (eg = –OCH$_2$CH$_2$O–) was used as the model complex, the activation free energy for the CO_2 insertion was calculated to be 16 kcal mol^{-1}.

3.5 CO_2 insertion into a metal–hydride bond

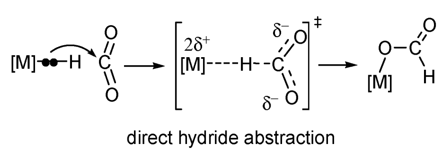
CO_2 insertion into an M–H bond can proceed *via* the normal insertion mechanism shown in Scheme 2. For example, as early as 1989, Sakaki and Ohkubo calculated the simplest example of CO_2 insertion into the (PH$_3$)$_2$Cu–H bond.[12] The activation barrier was calculated to be $\Delta E^\ddagger$ = 9.3 kcal mol^{-1} ($\Delta G^\ddagger \approx$ 19.3 kcal mol^{-1}). In 1997, Hutschka, Leitner and coworkers investigated theoretically CO_2 insertion into the (PH$_3$)$_2$Rh–H bond and found that CO_2 first coordinates to the metal center through η^2(C, O) side-on coordination mode and then a normal insertion occurs.[24] The activation barrier was calculated to be $\Delta E^\ddagger$ = 4.3 kcal mol^{-1} ($\Delta G^\ddagger \approx$ 14.3 kcal mol^{-1}). In 2011, Huang, Guan and coworkers studied theoretically CO_2 insertion into the Ni–H bond in the 16e metal complex (PCP)Ni–H (PCP = 2,6-C$_6$H$_3$(OP^{t}Bu$_2$)$_2$) and found that the insertion proceeds also *via* the normal insertion mechanism.[25]

A precondition for CO_2 insertion *via* the normal insertion mechanism shown in Scheme 2 is that the metal complexes to react with CO_2 should be coordination unsaturated. In the three examples discussed here, each of the metal complexes has vacant site(s) for CO_2 coordination. Metal complexes, which have already had an 18-electron configuration, can create vacant site(s) *via* ligand dissociation as the first event for CO_2 insertion if the ligand dissociation is to occur easily. In 2002, Musashi and Sakaki investigated theoretically rhodium-catalysed hydrogenation of CO_2.[26] This theoretical investigation showed that CO_2 insertion into *cis*-[RhH$_2$(PH$_3$)$_2$(L)$_2$]$^+$ (L = PH$_3$ or H$_2$O) proceeds *via* dissociation followed by the normal insertion mechanism. The overall activation electronic energies (considering both the ligand dissociation and the insertion steps) were calculated to be 65.8 and 42.6 kcal mol^{-1} for L = PH$_3$ and H$_2$O, respectively. The ligand dissociation electronic energies alone were 26.6 and 25.0 kcal mol^{-1} for L = PH$_3$ and CO, respectively.

Due to the spherical 1s valence orbital of H, a direct hydride abstraction by CO_2 is possible for insertion of CO_2 into an M–H bond (shown in Scheme 7). This mechanism does not need vacant site(s). When ligand dissociation is not to occur easily, this mechanism would dominate during insertion of CO_2 into a metal–hydride bond. In 2000, Sakaki and coworkers investigated theoretically ruthenium-catalysed hydrogenation of CO_2.[27] They compared the two mechanisms, normal insertion and direct abstraction, for reaction of CO_2 with the 18-electron model complex *cis*-RuH$_2$(PH$_3$)$_4$. It was found that the overall activation barrier *via* dissociation followed by a normal insertion ($\Delta E^\ddagger$ = 30.5 kcal mol^{-1}) is similar to that *via* the direct abstraction ($\Delta E^\ddagger$ = 29.3 kcal mol^{-1}). If bulkier ligands such as the experimental phosphines PMe$_3$ and PPh$_3$ were employed in the calculations, we expected that direct abstraction would be clearly favorable due to the steric effects. In 2011, Jun and Yoshizawa studied theoretically insertion reaction of CO_2 with the 18-electron complex (PNP)IrH$_3$ (PNP = 2,6-(diisopropylphosphino-methyl)-pyridine).[28] Ligand dissociation is difficult to occur



Scheme 6



Scheme 7

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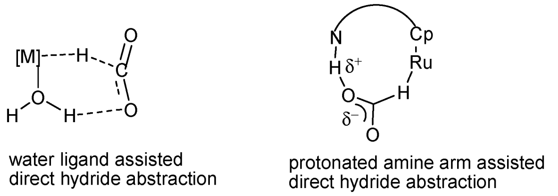
here because of the chelating coordination. Without a ligand dissociation, the normal insertion was calculated to have an activation barrier of $\Delta E^{\ddagger}$ = 41.0 kcal mol^{-1}. In a direct abstraction path, the activation barrier was calculated to be $\Delta E^{\ddagger}$ = 11.0 kcal mol^{-1} ($\Delta G^{\ddagger} \approx$ 21.0 kcal mol^{-1}). In 2007, Urakawa, Baiker and coworkers investigated theoretically hydrogenation of carbon dioxide catalysed by the 18-electron ruthenium dihydride complex *trans*-(dmpe)$_2$Ru(H)$_2$.[29] The CO_2 insertion into one of the Ru–H bonds was found to proceed *via* a direct hydride abstraction path with an activation barrier of $\Delta E^{\ddagger}$ = 9.5 kcal mol^{-1} ($\Delta G^{\ddagger} \approx$ 19.5 kcal mol^{-1}). In 2011, Yang investigated theoretically hydrogenation of carbon dioxide catalysed by PNP pincer iridium, iron and cobalt complexes. Direct hydride abstraction was found to be favorable for reactions of CO_2 with (PNP)IrH$_3$, (PNP)Fe(H)$_2$CO and (PNP)CoH$_3$ (PNP = 2,6-bis(di-iso-propylphosphinomethyl)pyridine).[30]

In the transition state structure for a direct hydride abstraction mechanism, accumulation of negative charge on the CO_2 oxygen atoms occurs (Scheme 7). Water molecules have been found to promote the abstraction process through hydrogen bonding with the CO_2 oxygen atoms in the transition state structure. In 2001, we investigated theoretically the promoting effect of water in ruthenium-catalysed hydrogenation of CO_2 into formic acid.[31] Without the involvement of water molecules, the activation free energy of CO_2 insertion into the Ru–H bond in TpRu(PH$_3$)(CH$_3$CN)(H) was calculated to be 31.8 kcal mol^{-1}. With a water molecule hydrogen-bonding with the CO_2 oxygen atoms, the activation free energy was calculated to be 26.2 kcal mol^{-1}.

In 2005 and 2006, Sakaki, Ohnishi and coworkers found in their theoretical studies that when water is used as a ligand, the promoting effect through hydrogen bonding is also evident.[32] A transition state structure is shown on the left-hand side of Scheme 8, illustrating the hydrogen-bonding interactions. The activation free energy of CO_2 insertion into one of the Ru–H bonds in the 16-electron complex *cis*-Ru(H)$_2$(PMe$_3$)$_3$(H$_2$O) *via* the direct hydride abstraction mode was calculated to be 8.4 kcal mol^{-1}, which is much smaller than that (21.5 kcal mol^{-1}) of CO_2 insertion into *cis*-Ru(H)$_2$(PMe$_3$)$_3$ *via* the normal insertion mechanism calculated at the MP4(SDQ) level of theory. In 2000, Matsubara studied theoretically insertion reaction of CO_2 with the 18-electron complexes ((η^5-C$_5$H$_4$)(CH$_2$)$_n$NMe$_2$H$^+$)Ru(dppm)(H) (n = 2 ,3) and found that hydrogen bonding of the protonated amine arm with one of the two CO_2 oxygens significantly promotes the insertion reaction (the transition state structure shown on the right-hand side of Scheme 8).[33] The activation barrier for the insertion was calculated to be $\Delta E^{\ddagger}$ = 6.2 kcal mol^{-1} ($\Delta G^{\ddagger} \approx$ 16.2 kcal mol^{-1}).



Scheme 8

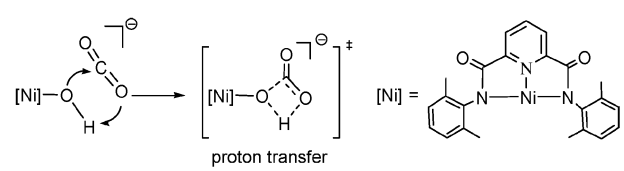
4. CO_2 insertion into an M–X bond

The insertion of CO_2 into an M–X bond (X = hydroxide, alkoxide, amide, *etc.*) has been widely investigated experimentally. For example, CO_2 has been found to insert into various M–OR bonds, where M includes titanium,[34] zirconium,[34] niobium,[34] chromium,[35] tungsten,[35] manganese,[36] rhenium,[36,37] iron,[34] palladium,[38,39] platinum,[40] copper[12a,41] *etc.*[42] Insertion of CO_2 into various metal–amide bonds was also reported, where the metals include titanium,[43] zirconium,[44] niobium,[45] tantalum,[45] chromium,[46] rhenium,[47] ruthenium,[48] zinc,[49] copper[50] *etc.*[42] However, there are only limited theoretical studies on the mechanism of CO_2 insertion into an M–X bond (X = hydroxide, alkoxide, amide, *etc.*). These limited theoretical studies showed that X has lone pair(s) of electrons which can help to activate CO_2. With the presence of the lone pair(s) of electrons, X in the M–X bond nucleophilically attacks the CO_2 C center to achieve the CO_2 activation.

In 2011, Lee, Rybak-Akimova, Holm and their coworkers studied experimentally CO_2 insertion into the Ni–OH bond in the 16-electron complex [(NNN)Ni(OH)]$^-$ (NNN = 2,6-pyridinedicarboxamidate).[11] Using a stopped-flow spectrophotometry technique, they determined the kinetics of the insertion reaction. In the study, they also carried out DFT calculations to investigate the reaction mechanism and compared the activation energies of a normal insertion mechanism and a mechanism that is initiated by the lone pairs of electrons of the OH$^-$ ligand nucleophilically attacking the CO_2 C center followed by migration of the OH$^-$ proton to one of the CO_2 oxygens (a proton transfer mechanism) as shown in Scheme 9.[11a] On the basis of their calculation results, the normal insertion mechanism has a much lower barrier ($\Delta H^{\ddagger}$ = 7.6 kcal mol^{-1}) than the proton transfer mechanism ($\Delta H^{\ddagger}$ = 21.5 kcal mol^{-1}).

In 1995, Sakaki and coworkers studied theoretically CO_2 insertion into L$_2$Cu–OH bonds (L = PH$_3$) *via* a normal insertion mechanism. They found that CO_2 insertion into the Cu–OH bond has almost no barrier and in the transition state the lone pair p orbital of the OH$^-$ ligand has strong bonding interaction with one of the CO_2 π^* orbitals.[12a] In 2006, Pápai, Schubert and coworkers investigated theoretically the reaction mechanism of the direct carboxylation of methanol to dimethylcarbonate catalysed by [Nb(OMe)$_5$]$_2$. They also found in their theoretical calculations a fast CO_2 insertion into one of the Nb–OMe bonds is *via* a normal insertion mechanism with an activation barrier of $\Delta E^{\ddagger}$ = 5.4 kcal mol^{-1} ($\Delta G^{\ddagger} \approx$ 15.4 kcal mol^{-1}).[51]

In 2012, Nova, Hazari and coworkers synthesized the PNP-supported nickel complexes (PCP)NiX (X = NH$_2$, OH; PCP = 2,6-C$_6$H$_3$(CH$_2$PMe$_2$)$_2$) and investigated CO_2 insertion into the Ni–X bond.[15] The mechanism of the CO_2 insertion



Scheme 9

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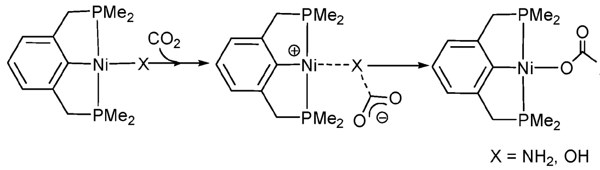


Scheme 10

was also studied with the aid of DFT calculations (Scheme 10). The DFT results show that the ligand X nucleophilically attacks CO_2 carbon to generate a zwitterionic species followed by a rearrangement to the insertion product. The rearrangement was calculated to be the rate-determining step ($\Delta G^{\ddagger}(NH_2)$ = 9.6 kcal mol^{-1}, $\Delta G^{\ddagger}(OH)$ = 12.2 kcal mol^{-1}).

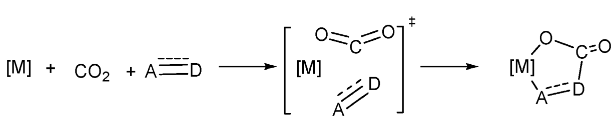
5. Oxidative coupling with unsaturated organic substrates

Oxidative coupling of CO_2 with olefins, alkynes and other unsaturated organic compounds mediated by a transition metal is another important fundamental step in the activation of CO_2 (Scheme 11). Coupling of CO_2 with olefins and alkynes mediated by Ni(0) complexes has been extensively studied theoretically. Ni(0) complexes containing olefin(s) or alkyne(s) as ligand(s) are typically three-coordinate with a planar geometry around the metal center. For example, a few spectroscopically characterized three-coordinate nickel complexes such as $(PR_3)_2Ni(C_2H_2)$, $Ni(PH_3)(C_2H_4)(C_2H_2)$, and $Ni(PR_3)(C_2H_2)_2$ (R = Me, Et, ^{i}Pr, Ch, Ph) have been reported.[52]

5.1 Oxidative coupling with alkynes

In early 1993, Sakaki, Arai and coworkers studied Ni-mediated oxidative coupling of CO_2 with acetylene theoretically.[53] Using the three- and four-coordinate model complexes, $Ni(PH_3)(C_2H_2)(CO_2)$ and $Ni(PH_3)_2(C_2H_2)(CO_2)$, they found that the oxidative coupling in the three-coordinate complex has a lower activation barrier ($\Delta E^{\ddagger}$ = 30 kcal mol^{-1}) than that ($\Delta E^{\ddagger}$ = 35 kcal mol^{-1}) in the four-coordinate complex with the aid of SD-CI calculations.[53]

In 1995, the same group studied the regioselectivity in the oxidative coupling of CO_2 with hydroxyacetylene.[54] Using the model complex $Ni(PH_3)(HC{\equiv}COH)(CO_2)$, they found from their theoretical calculations that the CO_2 carbon preferentially couples with the HC≡ carbon rather than the ≡COH carbon. The activation electronic energies were calculated to be 27 and 38 kcal mol^{-1}, respectively, for the two modes of oxidative coupling. This result is consistent with the notion that during the coupling process the CO_2 carbon acts as an electrophile. The HC≡ carbon is π-electron-richer than the ≡COH carbon and attacks the CO_2 carbon more readily. Interestingly, the reaction through the more favored coupling site is also more exothermic (ΔE = −31 kcal mol^{-1}) than that through the less favored coupling site (ΔE = −25 kcal mol^{-1}).



Scheme 11

In 2008, we carried out theoretical studies on coupling reactions of CO_2 with alkynes in $L_2Ni(\eta^2\text{-alkyne})$ (L = 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU)) complexes.[55] We considered two reaction pathways, associative and dissociative pathways. The associative pathway involves a direct coupling of a free CO_2 with the coordinated alkyne (CO_2 + $L_2Ni(\eta^2\text{-alkyne})$) to give a square planar coupling product. The dissociative pathway involves first dissociation of one L ligand followed by η^2-coordination of CO_2, then oxidative coupling of CO_2 with the alkyne ligand, and finally re-coordination of the L ligand. The theoretical calculations indicated that the associative pathway has lower activation free energies (by 2–3 kcal mol^{-1}) than the dissociative pathway. The coupling reactions of CO_2 with three different coordinated alkynes HC≡CMe, HC≡COMe and HC≡CCN were found to give the same regioselectivity although the three alkynes have very different electronic properties. The coupling was found to occur through the orbital interaction between the $\pi_{\perp}$ orbital of the η^2-coordinated alkyne and the CO_2 π^* orbitals. The nature of this type of orbital interactions explains the results that HC≡CCN gives the highest barrier (25.3 kcal mol^{-1}) while HC≡C(OMe) gives the lowest barrier (17.1 kcal mol^{-1}). The OMe substituent makes the π bonds more electron-rich while the CN substituent makes the π bonds electron-poor. Regarding the regioselectivity observed in coupling reactions of CO_2, steric repulsive interaction between the substituent at the alkyne carbon, which forms the new carbon–carbon bond with the CO_2 carbon, and the non-reacted C=O group of CO_2 was found to dominate.

In 2008, Graham, Buntine and coworkers also investigated the regioselectivity of the coupling of CO_2 with acetylene mediated by $(mDBU)_2Ni$ (mDBU = 1,4,5,6-tetrahydro-1,2-dimethylpyrimidine) using the DFT method.[56] The results obtained from this study are the same as those described above.

5.2 Oxidative coupling with alkenes

In 2004, Pápai, Aresta and coworkers studied theoretically the mechanism of coupling of CO_2 with C_2H_4 mediated by (bpy)Ni (bpy = 2,2′-bipyridine).[57] They found that coupling of CO_2 with C_2H_4 occurs *via* a single step by the approach of CO_2 on $(bpy)Ni(C_2H_4)$ with an activation barrier of $\Delta E^{\ddagger}$ = 19.0 kcal mol^{-1} ($\Delta G^{\ddagger} \approx$ 29.0 kcal mol^{-1}). In 2007, Graham, Buntine and coworkers investigated theoretically production of acrylic acid through $Ni(DBU)_2$-mediated coupling of CO_2 with ethylene.[58] Using a model DBU ligand (DBU = 1,4,5,6-tetrahydro-1,2-dimethylpyrimidine), the activation free energy for coupling of CO_2 with ethylene *via* approach of CO_2 on $(mDBU)_2Ni(C_2H_4)$was calculated to be 28.1 kcal mol^{-1}.

Coupling of CO_2 with ethylene mediated by *trans*-$M(C_2H_4)_2(PMe_3)_4$ (M = Mo, W) complexes has also been studied experimentally and theoretically. Experimentally, it was found that reaction of CO_2 with *trans*-$M(C_2H_4)_2(PMe_3)_4$ (M = Mo, W) gave dimeric products (eqn (2)).[59] *trans*-$M(C_2H_4)_2(PMe_3)_4$ (M = Mo, W) complexes are 18-electron complexes. It was assumed that they first dissociate one PMe_3 ligand to allow CO_2 to coordinate to the metal center. This dissociative mechanism is consistent with the following experimental findings. *trans*, *mer*-$[M(C_2H_4)_2(CO)(PMe_3)_3]$

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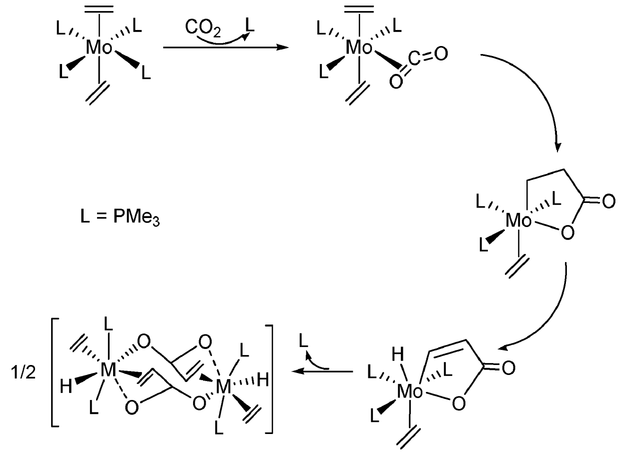
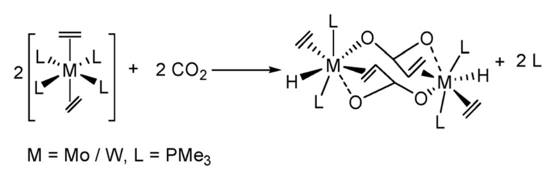


Scheme 12

(M = Mo, W) can be synthesized from the reaction of *trans*-[M(C_2H_4)$_2$(PMe_3)$_4$] and CO.[60,61]

In 2003, Schuber and Pápai investigated theoretically the mechanism for the reaction shown in eqn (2).[62] Scheme 12 shows the steps calculated for the formation of the dimeric product for the *trans*-Mo(C_2H_4)$_2$(PMe_3)$_4$ complex. A ligand substitution of CO_2 for PMe_3 occurs first followed by oxidative coupling in *trans*-Mo(C_2H_4)$_2$(PMe_3)$_3$(CO_2). This ligand substitution process was calculated to have a reaction electronic energy of −4.3 kcal mol^{-1}. Starting from *trans*-Mo(C_2H_4)$_2$-(PMe_3)$_3$(η^2-CO_2), the oxidative coupling was calculated to have an activation barrier of $\Delta E^{\ddagger}$ = 11.2 kcal mol^{-1} (Scheme 12). From the species derived from the oxidative coupling, β-hydrogen elimination occurs followed by dimerization to give the final product.

 (2)

6. Metal-mediated stoichiometric reactions

Transition metals or metal complexes are well known to be able to mediate reactions of CO_2. The reactions of CO_2, which were theoretically studied, can be categorized into the following categories: oxygen abstraction reactions ([M] + CO_2 → [M]O + CO), reductive disproportionation reactions (2 CO_2 + 2e$^-$ → CO_3^{2-} + CO), and metal-mediated reactions of CO_2 with other organic substrates.

6.1 Oxygen abstraction reactions

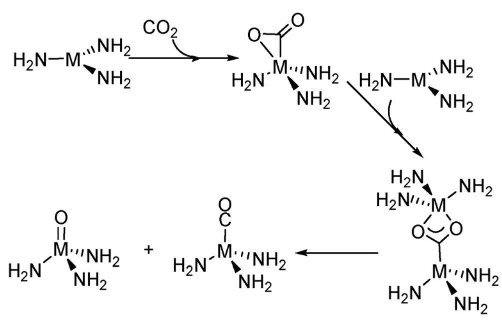
It has been shown in experiments that laser-ablated early transition metals (Sc, Ti, V, Zr, Nb, W, Ru, Mo) and some late transition metals or metal cations react with CO_2 to form metal oxides and CO.[63] Theoretical studies of these oxygen abstraction reactions showed that an insertion–elimination mechanism is operative (eqn (3)).[64] A metal atom (or cation) inserts and breaks one of the two C═O double bonds followed by elimination of CO. This mechanism can be easily understood. All of the metals (or metal cations) studied contain metal d electrons. Therefore, in the insertion step, occupied metal d orbitals can strongly interact with one π* orbital of CO_2 *via* an η^2-coordination mode. This type of metal(d)-to-CO_2(π*) back-bonding interaction weakens and facilitates the C═O bond breaking. It was found that the energy barrier of the insertion step is much smaller for an early transition metal than a late transition metal. Many early transition metals spontaneously insert into CO_2 with almost no barriers such as Sc, Ti, V, *etc.* However, for late transition metals, such as Ni, the energy barrier of the insertion step is substantially high (34.6 kcal mol^{-1}).[64o] This is easily understood because early transition metals are more inclined to transfer d electrons to the π* orbital of CO_2 than late transition metals.

 (3)

Experimentally, it was found that the Laplaza–Cummins complex Mo[N(^{t}Bu)Ar]$_3$ is active towards CS_2 but inactive towards CO_2.[65] In order to understand the reactivities of the complex, Ariafard, Yates and coworkers systematically carried out theoretical studies on reactions of M[N(^{t}Bu)Ar]$_3$ (M = Mo, Ta, Nb, V) with CO_2 shown in eqn (4).[66] The pathway examined in this theoretical work is shown in Scheme 13. The pathway considers that CO_2 first coordinates to the metal center *via* an η^2(C,O) side-on coordination mode followed by binding of a second metal complex to the two oxygen atoms of the coordinated CO_2 from which one C═O bond breaks to form a metal oxide and a metal carbonyl. It was found that the interaction between CO_2 and the three coordinate Mo[N(R)Ar]$_3$ having a d^3 quartet ground state is highly repulsive due to a significant mismatch in orbital energies between the CO_2 π* and the occupied metal d orbitals. Although the doublet Mo[N(R)Ar]$_3$ can interact strongly with CO_2, a highly significant barrier was calculated due to spin pairing (leading to a doublet state), steric bulk of the peripheral ligands and the entropy effect.[66a]

$$2\ \mathrm{M[N(tBu)Ar]_3} + \mathrm{CO_2} \longrightarrow \mathrm{OM[N(tBu)Ar]_3} + \mathrm{(CO)M[N(tBu)Ar]_3} \qquad (4)$$

When the metal center in the Laplaza–Cummins complex is changed to one of the group 5 metals, three-coordinate analogous complexes having a d^2 triplet ground state are



Scheme 13

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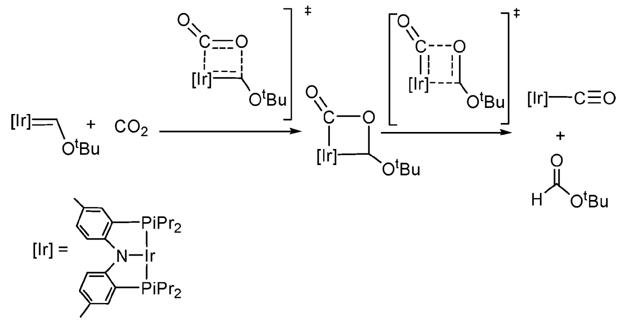


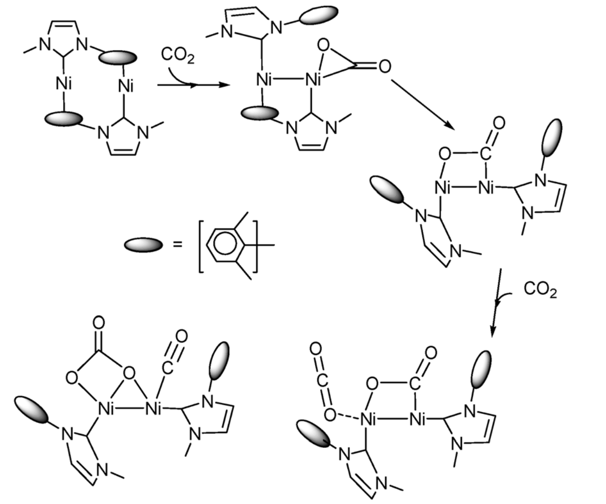
Scheme 14

derived. Theoretical calculations on the reactions of CO_2 with these analogous complexes by Ariafard, Yates and coworkers give rise to the following findings. Spin crossing leading to a singlet state is necessary for these complexes to interact efficiently with CO_2. Furthermore, the tantalum analogue can successfully activate CO_2. The niobium analogue will likely coordinate CO_2 but may not proceed further. The vanadium analogue has a significant initial coordination barrier and is not expected to activate CO_2.[66b] The ability of the tantalum analogue to activate CO_2 is related to the fact that tantalum has much more diffuse d orbitals and the tantalum d^2 metal center has smaller spin pairing barrier than the corresponding niobium and vanadium metal centers.

In 2008, Whited and Grubbs found experimentally that CO_2 reacts with an iridium carbene to form iridium carbonyl shown in Scheme 14.[67] Interestingly, the carbene ligand, not the metal center, acts as an oxygen abstractor in this reaction. In 2009, Ariafard, Yates and coworkers carried out DFT study on the mechanism of cleavage of CO_2 by the iridium Fischer carbene.[68] The DFT study showed that reaction proceeds *via* an oxygen-atom metathesis from CO_2 to the metal-bound Fischer carbene. CO_2 is activated *via* interaction between the occupied d_{z^2} metal orbital and one unoccupied π^* orbital of CO_2.

6.2 Reductive disproportionation reactions

Recently, Sadighi and co-workers reported their experimental finding of the reduction of CO_2 by the dinuclear complex $[(IPr)Ni(0)]_2$ (where IPr = 1,3-bis(2,6-diisopropylphenyl)-imidazole-2-ylidene) shown in eqn (5).[69] The net reaction related to CO_2 can be considered as a reductive disproportionation reaction ($2\ CO_2 + 2e^- \rightarrow CO_3^{2-} + CO$) in which the two electrons are provided by Ni(0) (being oxidized to Ni^{2+}). Based on this experimental work, we carried out theoretical studies on this reaction and were able to propose a reasonable mechanism to account for the reaction.[70] Scheme 15 illustrated the reaction mechanism proposed. In the calculations, non-coordinated 2,6-diisopropylphenyl substituents of the IPr ligands were modeled by methyl groups. For the metal-coordinated 2,6-diisopropylphenyl rings, the isopropyl groups were simplified to methyl groups. The binuclear complex $[(NHC)Ni]_2$ has an electron count of 36, indicating that each metal center is saturated. From the binuclear complex, a ligand exchange gives an η^2-CO_2-coordinated intermediate which has an electron count of 32. In the η^2-CO_2-coordinated intermediate, a formal Ni-to-Ni dative bond exists when the η^6-arene bound Ni center is viewed as an 18e center and the



Scheme 15

CO_2-coordinated Ni center satisfies the 16-electron rule. From the intermediate, dissociation of the second arene ring accompanied by the migration of the metal-bonded O of the coordinated CO_2 from one Ni center to another Ni center gives a new intermediate in which each metal has a T-shape coordination geometry and there is formally a metal–metal single covalent bond between two Ni(I) metal centers. This step was found to have an activation free energy of 29.0 kcal mol^{-1} and to be endergonic by 3.3 kcal mol^{-1}. The new intermediate is coordinatively unsaturated and is capable of coordinating another carbon dioxide molecule to allow an oxygen transfer, which resembles the head-to-tail coupling step proposed earlier by Cooper, Kubiak and their coworkers,[71,72] to occur. The second CO_2 coordination together with the oxygen transfer requires an overall activation free energy of 27.3 kcal mol^{-1}. Completion of the oxygen transfer gives a complex containing both CO and CO_3^{2-} ligands, completing the reductive disproportionation reaction of $2CO_2 + 2e^- \rightarrow CO$ and CO_3^{2-}. From the complex containing both CO and CO_3^{2-} ligands, the products shown in eqn (5) can be easily derived. Based on calculation results, we found that the Ni centers in the dinuclear complex are electron-rich and the back donation of d electrons from metal center to π^* of CO_2 facilitates the breaking of one of the two C=O bonds.

$$1.5\ [(IPr)Ni]_2 + 3\ CO_2 \longrightarrow [(IPr)Ni]_2(\mu\text{-}CO)(\mu\text{-}\eta^2,\eta^2\text{-}CO_2) + [(IPr)NiCO_3] \quad (5)$$

$$1.25\ L^{t}Bu{-}Fe{-}N_2{-}Fe{-}L^{t}Bu + 2\ CO_2 \longrightarrow$$
$$0.25\ L^{t}Bu{-}Fe{-}(CO)_2 + 1.25\ L^{t}Bu{-}Fe{-}CO_3{-}Fe{-}L^{t}Bu + N_2$$
$$L^{t}Bu = \text{2,2,6,6-tetramethyl-3,5-bis[(2,6-diisopropylphenyl)-imino]hept-4-yl} \quad (6)$$

In 2010, Ariafard, Yates and coworkers carried out DFT study on the mechanism of the reductive cleavage of CO_2 by a bis-β-diketoiminatediiron dinitrogen complex.[73] The theoretical study is based on the experimental observation that CO_2 reacts with $L^{tBu}Fe$–N_2–FeL^{tBu} (L^{tBu} = 2,2,6,6-tetramethyl-3,5-bis[(2,6-diisopropylphenyl)-imino]hept-4-yl), affording dicarbonyliron complex $L^{tBu}Fe(CO)_2$, carbonatodiiron complex and dinitrogen shown in eqn (6)).[74] In the calculations, L^{tBu}

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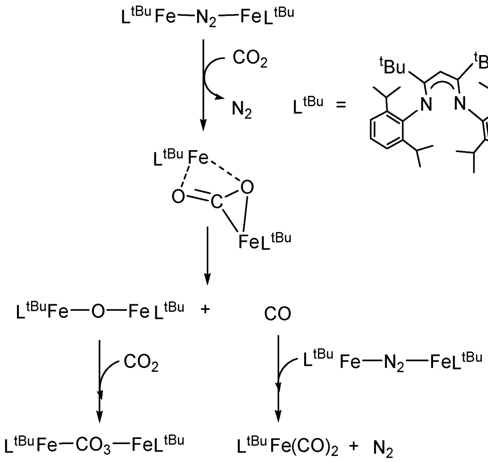


Scheme 16

was modeled by L = HC(CHNMe)$_2$. The mechanism shown in Scheme 16 was proposed to account for the reaction. Substitution of CO_2 for N_2 occurs as the first step. The calculation results indicate that this step is rate-determining with a free energy barrier of 21.6 kcal mol^{-1}. Next, reductive cleavage of CO_2 gives LFeOFeL + CO. Further reaction of LFeOFeL with a second CO_2 to form the product LFe–CO_3–FeL. Formation of LFe(CO)$_2$ is *via* reaction of the reactant LFe–N_2–FeL with CO produced from the reductive cleavage step.

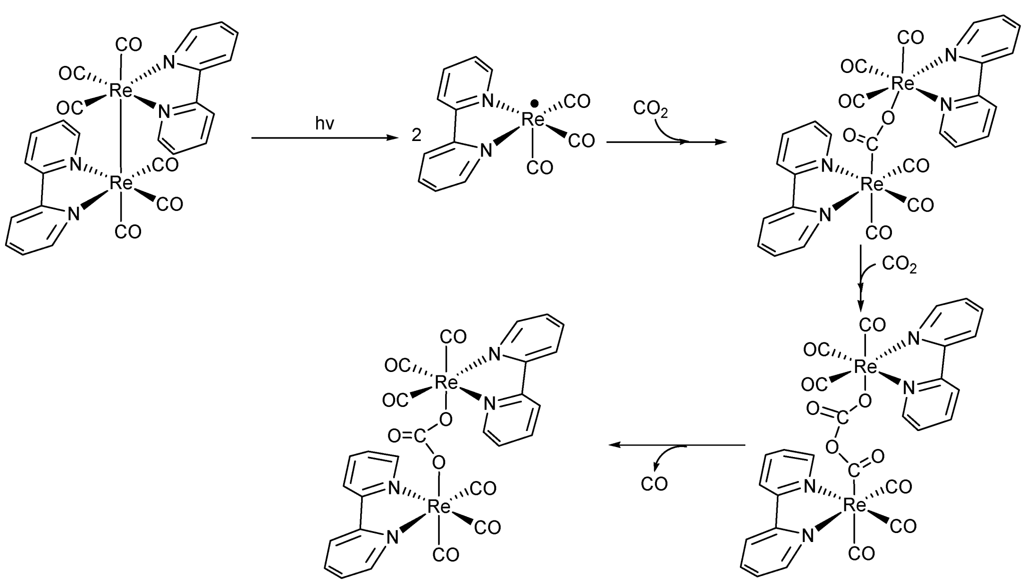
Experimentally, Meyer, Fujita and their coworkers investigated CO_2 reduction mediated by a rhenium dimer complex and observed several products including CO and a rhenium carbonate dimer.[75] In 2012, Muckerman and coworkers studied theoretically the mechanism of the CO_2 reduction.[76] The mechanism studied is shown in Scheme 17. The dimer complex breaks the Re–Re metal–metal bond to form a five coordinate 17-electron species as the first step. Then, two molecules of the 17-electron species react with one molecule of CO_2 to form a carboxylate dimer with a reaction enthalpy of −36.0 kcal mol^{-1}. Another molecule of CO_2 inserts into the Re–O bond in the carboxylate dimer with a reaction enthalpy of 7.8 kcal mol^{-1}. The insertion activation enthalpy was calculated to be 22.0 kcal mol^{-1}. The final step is the extrusion of CO, which is facile and exothermic by 1.9 kcal mol^{-1}. The overall reaction is exothermic by 30.1 kcal mol^{-1} and the rate-determining step is the CO_2 insertion.

6.3 Ta$^+$ (or W+) mediated reactions with CH_4

In 1995, Schwarz and coworkers investigated theoretically the possibility of using a transition-metal ion to mediate coupling reaction of CO_2 with methane and concluded that Ta^+ and W^+ were capable of mediating the reaction of $M^+ + CH_4 + 2CO_2 \rightarrow MO_2^+ + H_2 + CO + O{=}C{=}CH_2$ and making the reaction exothermic.[77] In 1998, Koch and coworkers carried out DFT study on the detailed mechanism of the coupling reaction (mediated by Ta^+).[78] The reaction was found to involve three major steps: (1) tantalum carbene is formed; (2) tantalum carbene reacts with CO_2 to form $O{=}Ta{=}CH_2^+$; (3) $O{=}Ta{=}CH_2^+$ reacts with a second CO_2 to form $O{=}Ta{=}O^+$ (shown in Scheme 18). The last step (a metathesis step) is the rate-determining step and the overall activation free energy was calculated to be 53.3 kcal mol^{-1}. The overall reaction is exergonic by 60.5 kcal mol^{-1}.

6.4 Niobium nitride-mediated reactions with MeC(O)X (X = Cl, OC(O)Me)

In 2010, Cummins and coworkers developed a synthetic cycle for reduction of CO_2 to CO, mediated by the anionic niobium nitride complex [NNb(N^{t}BuAr)$_3$]$^-$ (Ar = 3,5-$C_6H_3Me_2$) shown in Scheme 19.[79] This is another example of using an organic species as an oxygen abstractor in the reactions. First, the anionic nitride complex [Na][NNb(N^{t}BuAr)$_3$] (**1Nb**) reacts with CO_2 to give the carbamate complex [Na][O_2CNNb(N^{t}BuAr)$_3$] (**2Nb**). The resulting complex **2Nb**



Scheme 17

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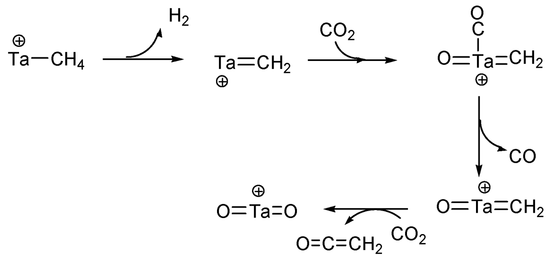
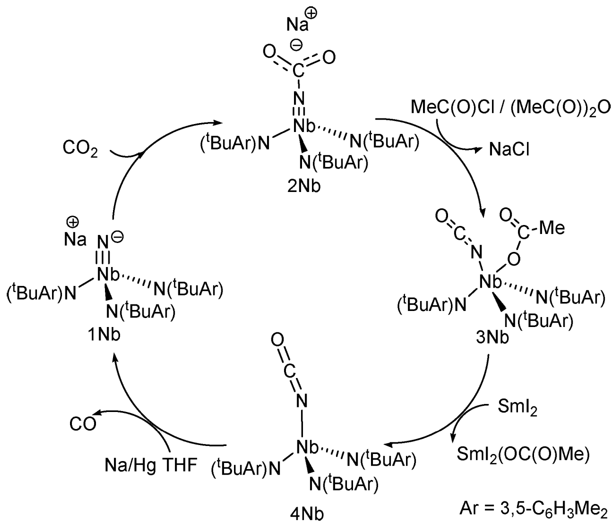


Scheme 18



Scheme 19

reacts with MeC(O)X (X = Cl or OC(O)Me) to give an isocyante–acetate complex [Na][OCNNb(OCOMe)(N^{t}BuAr)$_3$] (**3Nb**), which can be further reduced by SmI_2 to give isocyanate complex OCNNb(N^{t}BuAr)$_3$ (**4Nb**). Further reduction of **4Nb** by Na/Hg extrudes CO and regenerates **1Nb**.

In 2011, we carried out DFT studies on the reaction sequence shown in Scheme 19 in detail.[80] In the DFT calculations, [NNb(N^{t}BuAr)$_3$]$^-$ was modeled by [NNb(N^{t}BuMe)$_3$]$^-$. The initial binding of CO_2 to the terminal nitride nitrogen atom to give an anionic carbamate complex was found to be facile with an activation free energy of 6.2 kcal mol^{-1} and exergonic by 9.7 kcal mol^{-1}. Reaction of the anionic carbamate complex (**2Nb**) with MeC(O)Cl resembles a substitution of the anionic complex for chloride to give an intermediate, with a free energy barrier of 10.5 kcal mol^{-1}, formally having MeC(O)$^+$ bonded to one of the CO_2 oxygens in the anionic carbamate complex. The intermediate easily isomerizes to **3Nb** *via* cleavage of a C–O single bond and simultaneous formation of a new Nb–O bond. **2Nb** + MeC(O)Cl → **3Nb** + Cl$^-$ is exergonic by 37.0 kcal mol^{-1}. Two other research groups also independently studied **1Nb** + CO_2 → **2Nb** and **2Nb** + MeC(O)X → **3Nb** + X$^-$. Wang and coworkers compared the reactivity of NM(N^{t}BuAr)$_3^-$ towards CO_2 and found the reactivity order of M = V < Nb < Ta.[81] Angelis and coworkers emphasized the role of the Na$^+$ counterion in the choice of pathways for **2Nb** + MeC(O)X → **3Nb** + X$^-$.[82] It was found that in the presence of Na$^+$, a two-step process, which resembles what we found and described above, is responsible for the conversion. However, in the absence of Na$^+$, a one-step concerted process, which has a much higher barrier and involves simultaneously C–X and C–O bond breaking as well as Nb–O bond formation, was found to be responsible. Interestingly, in our DFT calculations without consideration of Na$^+$, we did not see the effect of the counter cation.[79] We believe that the different results are due to use of different models for MeC(O)X. MeC(O)Cl was used in our calculations while acetic anhydride was used in the work reported by Angelis and coworkers.

The last step shown in Scheme 19 is reduction of the isocyanate complex **4Nb** by Na to lead to cleavage of the C═N double bond in the isocyanate ligand and extrusion of CO. Our DFT calculations indicate that transfer of the valence electron from Na to **4Nb** generates a triplet Nb(III) then a singlet Nb(III) *via* an intersystem crossing. The singlet Nb(III) center can bind the isocyanate ligand and facilitate the C═N bond cleavage by formation of an Nb≡N bond. The CO extrusion process is exergonic by 16.6 kcal mol^{-1} and has a free energy barrier of 20.8 kcal mol^{-1}.

6.5 Nickel-mediated reactions with H_2

In 2005, Hu and coworkers reported their theoretical study on the mechanism of Ni-mediated reduction of CO_2 with H_2 to give CO and H_2O.[83] The reaction sequence shown in Scheme 20 was calculated. First, CO_2 and H_2 coordinate to Ni to form Ni(η^2-CO_2)(η^2-H_2). Then Ni inserts into the coordinated C–O bond of CO_2 followed by two consecutive H migration steps from the coordinated H_2 to the oxo ligand in the Ni=O moiety to form a H_2O ligand. Dissociation of H_2O and formation of the nickel carbonyl NiCO complete the Ni-mediated reduction reaction. The overall reaction is exothermic by 71.1 kcal mol^{-1} with an activation electronic energy of 32.9 kcal mol^{-1}. The rate-determining step is the insertion of Ni into one C–O bond of CO_2.

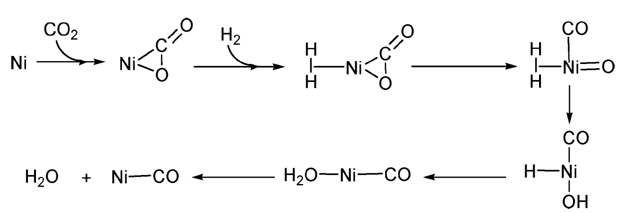
7. Metal-catalysed reactions

Reactions of CO_2 catalysed by transition metal complexes have also been extensively studied theoretically. In this section, we will summarize the reactions theoretically studied.

7.1 Reactions with organoboranes

$$CO_2 + B_2pin_2 \xrightarrow{\text{(NHC)Cu(Bpin)}} CO + (pinB)_2O \qquad (7)$$

In 2006, we, in collaboration with Professor Marder, investigated theoretically the reaction mechanism of the reduction of CO_2 to CO catalysed by copper(I) boryl complexes shown in eqn (7), which had been experimentally studied by Sadighi



Scheme 20

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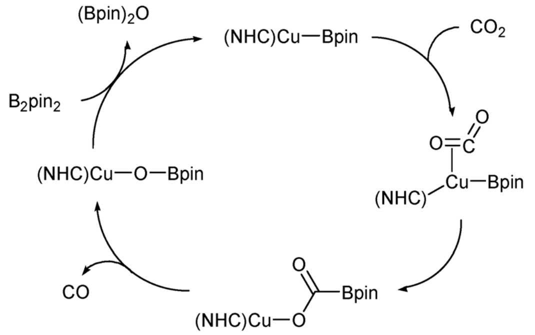
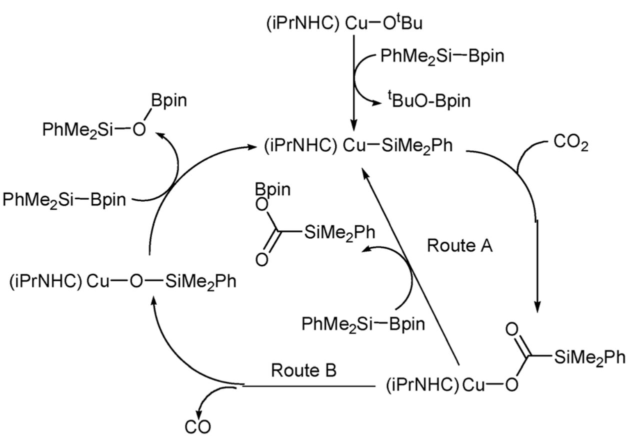


Scheme 21

et al.[23] On the basis of the theoretical studies, the mechanism shown in Scheme 21 was proposed, which composed of CO_2 insertion into a Cu–B bond to give a Cu–O–C–B linkage, and boryl migration from C to O to release CO, followed by a σ-bond metathesis between B_2pin_2 and (NHC)Cu(OBpin) (NHC = *N*-heterocyclic carbene) to regenerate the copper(I) boryl catalyst. The back-bonding interaction between the Cu–B σ-bond and the CO_2 carbon gives rise to a small CO_2 insertion free energy barrier (16.0 kcal mol^{-1}). The step of releasing CO is the rate-determining step with an activation free energy of 22.0 kcal mol^{-1}. The σ-bond metathesis step was calculated to have a relatively small free energy barrier (14.3 kcal mol^{-1}). The overall reaction free energy for eqn (7) is −38.0 kcal mol^{-1}.

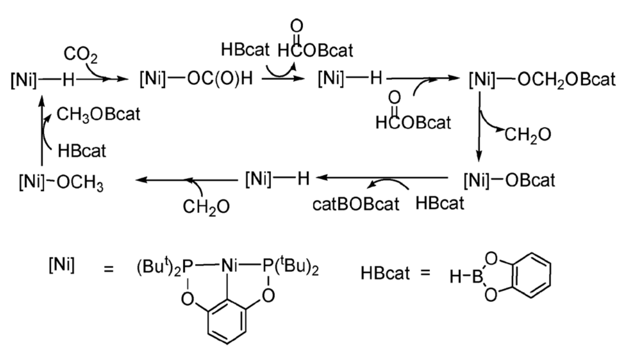
Recently, Kleeberg *et al.* found in their experimental work that copper-catalyzed reduction of CO_2 with $PhMe_2Si$–Bpin leads not only to CO and $PhMe_2Si$–O–Bpin, but also $PhMe_2$-Si–C(O)–OBpin as an additional reduction product.[84] In this experimental work, DFT calculations have also been carried out to shed light on the reaction mechanism. Scheme 22 shows the proposed reaction mechanism to account for the reaction. DFT calculations indicate that both the two routes shown in Scheme 22 are energetically feasible. The insertion of CO_2 into Cu–Si was calculated to be exergonic by 6.6 kcal mol^{-1} and to have a free energy barrier of 22.3 kcal mol^{-1}. In route A, the transmetallation to regenerate the Cu–Si active species requires a free energy barrier of 21.9 kcal mol^{-1}, but was found to be slightly endergonic (by 2.3 kcal mol^{-1}). The reaction free energy for this route (CO_2 + $PhMe_2Si$–Bpin → $PhMe_2Si$–C(O)–OBpin) was calculated to be −4.3 kcal mol^{-1} only. In route B, release of CO was found to have a high free energy barrier (31.3 kcal mol^{-1}) but is exergonic (by 10.8 kcal mol^{-1}). The last step, which is a transmetallation step, requires a free energy barrier of 22.0 kcal mol^{-1} and is exergonic by 18.5 kcal mol^{-1}. The reaction free energy of the CO_2 to CO reduction reaction (CO_2 + $PhMe_2Si$–Bpin → CO + $PhMe_2Si$–O–Bpin) is −35.9 kcal mol^{-1}.



Scheme 22

In 2010, Guan and coworkers found experimentally that a PCP-pincer nickel hydride is capable of catalysing reduction of CO_2 with HBcat to form a methoxy boron compound.[85] In 2011, they carried out a computational study on the mechanism.[86] The proposed mechanism for the catalysed reaction is shown in Scheme 23. Experimentally, [Ni]OOCH, HCOOBcat, CH_3OBcat and catBOBcat were identified except CH_2O in the catalysed reaction. The proposed overall reaction is CO_2 + 3 HBcat → catB–O–Bcat + H_3C–O–Bcat. The whole process is *via* an alternative hydride transfer and metathesis mechanism. The role of [Ni]H is to transfer HBcat hydride to CO_2 and carbonyl intermediates formed. There are three hydride transfers during the whole catalytic process. The rate-determining step is the hydride transfer to the intermediate HCOOBcat with an activation free energy of 32.5 kcal mol^{-1}.

In 2008, it was experimentally found that organoboranes, such as aryl- and alkenylboronate esters, can be carboxylated by reaction with CO_2 in the presence of the catalyst [(IPr)CuCl] (IPr = 1,3-bis(2,6-diisopropylphenyl)imidazol-2-ylidene) in refluxing THF (eqn (8)).[87] In 2010, we, in collaboration with Professor Marder, carried out DFT studies on the detailed mechanism of the catalysed reactions.[13] Our computational results based on the model complex (NHC)Cu(OMe) and substrate PhBneop showed that the catalysed carboxylation occurs *via* three major steps shown in Scheme 24: (1) transmetalation of PhBneop to LCuX assisted by base; (2) CO_2 insertion into the Cu–Ph bond to give the Cu–OC(O)Ph carboxylate intermediate; and finally, (3) reaction of this intermediate with the base adduct of the PhBneop substrate, which regenerates the catalyst and affords the benzoate product. Insertion of CO_2 into the (NHC)Cu–Ph was found to be the rate-determining step with an activation free energy of 19.6 kcal mol^{-1}. The overall reaction free energy is −56.3 kcal mol^{-1}. The insertion process proceeds *via* nucleophilic attack of the Cu–Ph π bond on the CO_2 carbon. Interestingly, we found that the transmetalation likely proceeds



Scheme 23

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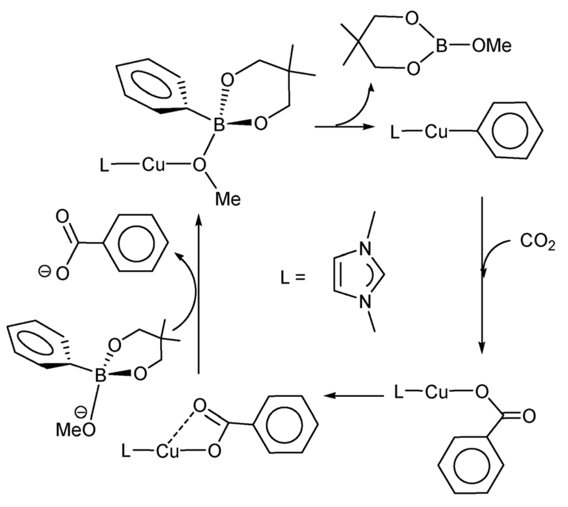
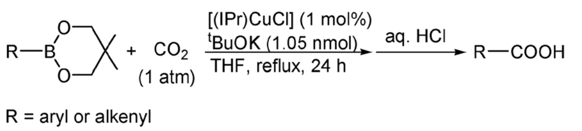


Scheme 24

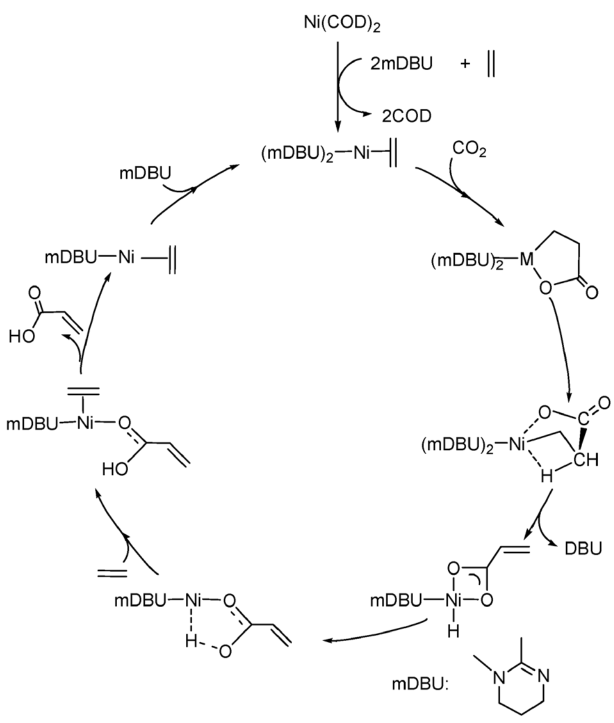
via formation of the ArBneopKOR adduct rather than by reaction of KOR with (NHC)CuCl.



(8)

7.2 Coupling reactions with ethylene

In 2007, Graham, Buntine and coworkers carried out a DFT study on the mechanism of catalytic coupling reaction of CO_2 with ethylene over a Ni(DBU)$_2$ catalyst.[58] The mechanism calculated is shown in Scheme 25. The first step is oxidative



Scheme 25

coupling of CO_2 with olefin with an activation solvation-corrected free energy of 24.6 kcal mol^{-1} and reaction free energy of -12.8 kcal mol^{-1}. Then a two-step process is related to the β-H elimination with an overall activation free energy of 39.3 kcal mol^{-1} and endergonic by 9.2 kcal mol^{-1}. From species derived from the β-H elimination, reductive elimination to form a new O–H bond is followed by coordination of an olefin molecule. The product molecule, acrylic acid, is then released. Finally, further coordination of an mDBU molecule regenerates the initial active species. The overall activation free energy for the whole catalytic cycle was calculated to be 39.3 kcal mol^{-1} from the initial active species to the transition state related to β-H elimination. The reaction free energy for the reaction of $CH_2{=}CH_2 + CO_2 \rightarrow CH_2{=}CHCOOH$ was calculated to be 10.2 kcal mol^{-1}.

7.3 Coupling reactions with epoxides

There are several theoretical studies on mechanism of coupling reactions of CO_2 with epoxides catalysed by metal complexes.[88] The main question here is whether to activate CO_2 or epoxides first. If the metal complexes contain ligands such as OR in [M]–OR having active lone pair(s) of electrons associated with O, CO_2 would be activated first because these active lone pair(s) of electrons can help activate CO_2. Otherwise, epoxides will be activated first by coordinating to the metal center.

In 2009, Wu and coworkers performed a theoretical study on the mechanism of Ni(0)-catalysed coupling of CO_2 with epoxyethane.[88a] The possible pathways (routes I, II and III) examined in the study are shown in Scheme 26. In route I, oxidative addition of epoxyethane forms an oxametallacyclobutane intermediate which lies 16.9 kcal mol^{-1} below the starting point with an activation free energy of 19.0 kcal mol^{-1}. Then CO_2 insertion into the Ni–O bond forms a metallacarbonate intermediate which lies 12.1 kcal mol^{-1} below the starting point with an activation free energy of 22.5 kcal mol^{-1} relative to oxametallacyclobutane intermediate. The last step is reductive elimination of cyclic carbonate, which is the rate-determining step. The overall activation free energy is 44.1 kcal mol^{-1}. In route II, ring-opening of epoxyethane takes place with cleavage of a C–O bond of epoxide, formation of new Ni–$O_{epoxide}$ and O_{CO2}–$C_{epoxide}$ bonds following a side-on CO_2 coordination, of which an activation free energy of 54.1 kcal mol^{-1} was calculated. The high activation energy excludes route II. In route III, the first step (oxidative addition of epoxyethane) is the same as that in route I. CO_2 inserts into the Ni–O bond followed by dissociation of PH_3 with an activation free energy of 23.9 kcal mol^{-1}. The last step is reductive elimination of cyclic carbonate, which is the rate-determining step. The overall activation free energy is 43.5 kcal mol^{-1}. The reaction free energy for $CH_2CH_2O + CO_2 \rightarrow OCH_2CH_2C(O)O$ was calculated to be -4.8 kcal mol^{-1}. Based on the results, it was concluded that routes I and III are competitive.

In 2009, Guo and coworkers reported a theoretical study on the mechanism of Cu(I) cyanomethyl catalysed coupling of propylene oxide with CO_2.[88b] The mechanism includes two major stages (Scheme 27). In stage 1, CO_2 insertion into a Cu–C bond affords a copper carboxylate intermediate followed by CO_2-assisted isomerization of a cyanomethyl ligand to a vinylideneamino ligand. A direct insertion of CO_2 into the Cu–C bond could not be found in the calculations.

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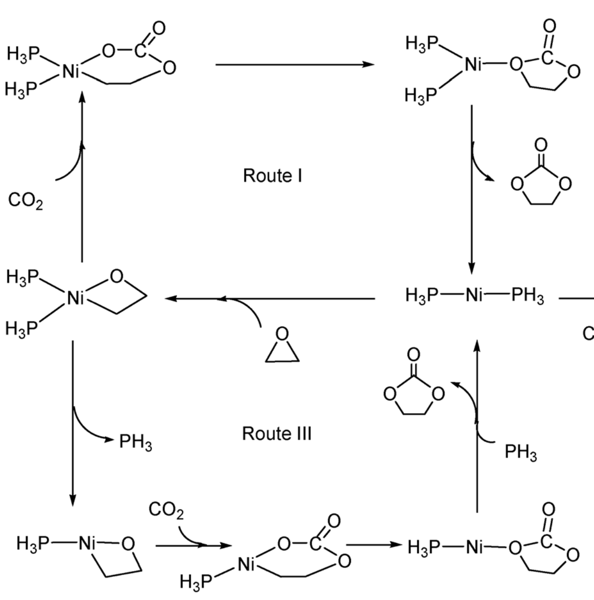
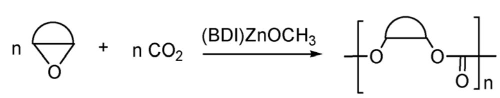


Scheme 26

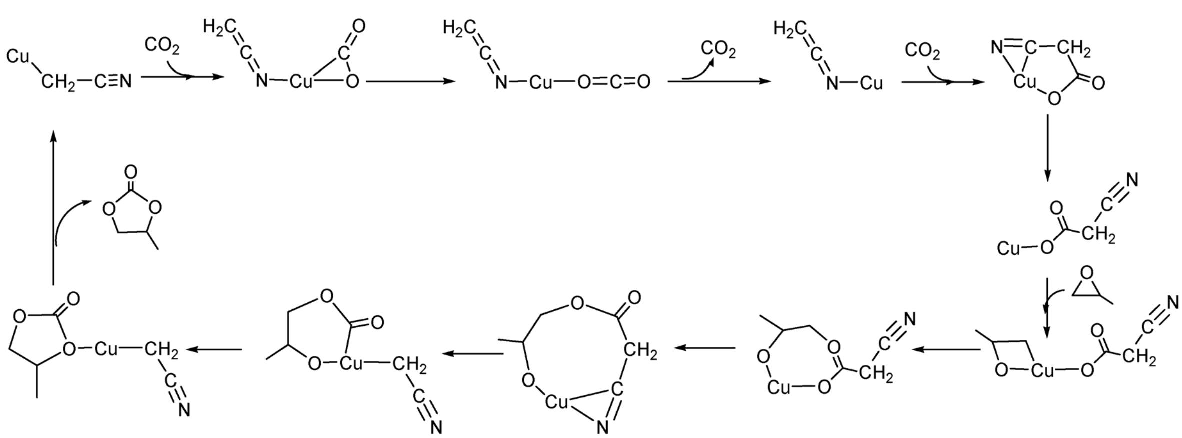
After coordination of CO_2, the copper cyanomethyl $N{\equiv}CCH_2Cu$ first isomerizes to $CH_2{=}C{=}NCu$ with an activation free energy of 23.1 kcal mol^{-1}. Through a five-membered ring intermediate, CO_2 inserts into the Cu–C bond. Isomerization from $N{\equiv}CCH_2Cu$ to $CH_2{=}C{=}NCu$ is the rate-determining step during this stage. In stage 2, oxidative addition of propylene oxide takes place, followed by a conversion of the four-membered ring intermediate to a seven-membered ring intermediate. Then through an eight-membered ring intermediate, the seven-membered ring intermediate rearranges to a six-membered metallacarbonate intermediate, which is the rate-determining step of this stage (also of the whole catalytic reaction) with an activation free energy of 60.3 kcal mol^{-1} relative to the four-membered ring intermediate. The last step is reductive elimination of cyclic carbonate. The reaction free energy for $CH_3CHCH_2O + CO_2 \rightarrow OCH_2CH(CH_3)OC(O)$ was calculated to be -17.4 kcal mol^{-1}. The overall activation barrier seems inaccessibly high in the proposed mechanistic pathway.

In 2002, Morokuma and coworkers reported a theoretical study on the mechanism of (BDI)ZnOCH$_3$-catalysed alternating copolymerization of CO_2 with epoxide (eqn (9)). Here, BDI [$=$N(2,6-^{i}Pr$_2$C$_6$H$_3$)C(Me)CHC(Me)N(2,6-^{i}Pr$_2$C$_6$H$_3$)] is a chelating β-diimine ligand and cyclohexane oxide was used as an epoxide substrate in the calculations.[88c] There are four possible insertions: (1) insertion of CO_2 into the Zn–OMe bond to form a metalla-carbonate intermediate; (2) insertion of epoxide into the Zn–OMe bond; (3) insertion of CO_2 into a Zn–carbonate bond; (4) insertion of epoxide into a Zn–carbonate bond. It was found that



(9)

CO_2 insertion into (BDI)Zn–OMe is kinetically more favorable (barrierless with a reaction free energy of -5.8 kcal mol^{-1}), although epoxide insertion into (BDI)Zn–OMe is thermodynamically more favorable (with an activation free energy of 12.8 kcal mol^{-1} and with a reaction free energy of *ca.* -40.0 kcal mol^{-1}). CO_2 inserts into the Zn–OMe bond much more easily than epoxide does. CO_2 insertion into a Zn–carbonate bond is endergonic by 10.4 kcal mol^{-1} with an activation free



Scheme 27

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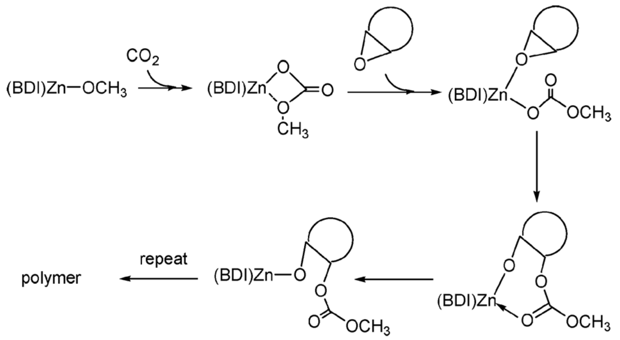


Scheme 28

energy of 10.9 kcal mol^{-1}. Therefore, the second CO_2 insertion into a Zn–carbonate bond is reversible with a reversible activation free energy of 0.5 kcal mol^{-1}, which prohibits formation of the CO_2 polymer. Epoxide insertion into a Zn–carbonate bond is exergonic by 32.0 kcal mol^{-1} with an activation free energy of 12.0 kcal mol^{-1}.

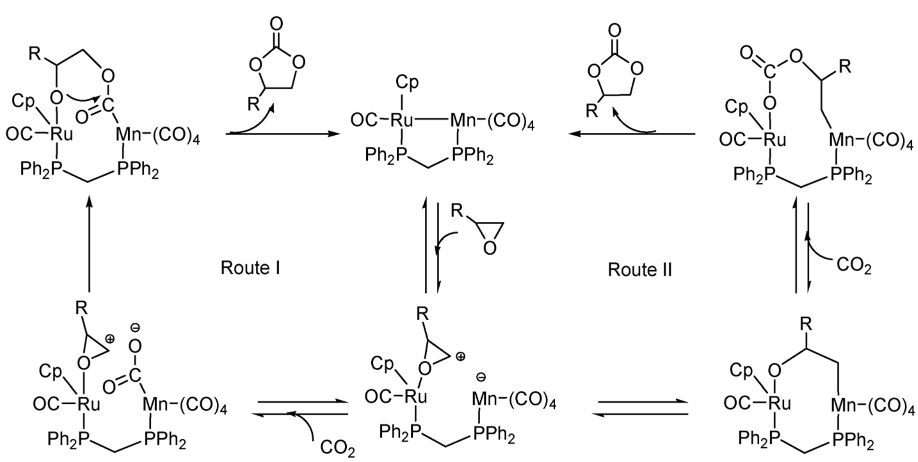
Due to the relatively small activation barrier and the strong thermal driving force, alternating copolymerization of CO_2 with epoxide is possible. The rate-determining step in the copolymerization reaction is the epoxide insertion into a Zn–carbonate bond. The favorable mechanism is given in Scheme 28.

In 2005, Qiu and coworkers reported in their experimental work that the transition-metal complex $Re(CO)_5Br$ catalysed the coupling reactions of CO_2 with epoxides to give cyclic carbonates under solvent-free conditions.[89] In 2010, Wu and coworkers investigated the reaction mechanism theoretically. In their calculations, chloromethyloxirane was used as a prototype of epoxides, considering the high conversion when it reacts with CO_2. The reaction mechanism is similar to route I of the Ni(0)-catalysed coupling of CO_2 with epoxyethane shown in Scheme 26. $Re(CO)_5Br$ first dissociates one CO to give the active catalyst specie $Re(CO)_4Br$. The mechanism can be divided into three main steps: (1) epoxide oxidative addition to form an oxametallacyclobutane intermediate; (2) CO_2 insertion; (3) reductive elimination of cyclic carbonate.

In 2006, Lau and coworkers showed experimentally that the heterobimetallic Ru–Mn complexes ([(η^5-C_5H_5)Ru(CO)-(μ-dppm)$Mn(CO)_4$], [(η^5-C_5Me_5)Ru(μ-dppm)(μ-CO)$_2$$Mn(CO)_3$], dppm = bidiphenylphosphinomethane) can catalyze the coupling reactions of epoxides with CO_2 to yield cyclic carbonates.[90] Because of the metal–metal bond polarity, one metal center can be used as a Lewis base to activate CO_2 while the other metal center can be used as a Lewis acid to activate epoxide. Two possible reaction pathways were proposed based on the complex ([(η^5-C_5H_5)Ru(CO)(μ-dppm)$Mn(CO)_4$] (Scheme 29). In collaboration with Lau, we carried out DFT calculations to examine the feasibility of the two reaction pathways.[90] In the DFT calculations, the phenyl groups in the dppm ligand of [(η^5-C_5H_5)Ru(CO)(μ-dppm)$Mn(CO)_4$] were replaced by hydrogen atoms. Both routes begin with heterolytic cleavage of the Ru–Mn bond followed by coordination of an epoxide molecule to the Lewis acidic Ru center to form a zwitterionic species in which the Mn center formally carries a negative charge while the Ru center formally carried a positive charge. In route I, the Lewis basic Mn center nucleophilically attacks the CO_2 carbon and activates CO_2 to form a metallocarboxylate anion, which then ring-opens the epoxide; ring-closure gives the cyclic carbonate. In route II, the Lewis basic Mn center ring-opens the Ru-attached epoxide to afford an alkoxide intermediate, CO_2 insertion into the Ru–O bond, followed by ring-closure, yields the product. Route II was calculated to be more favorable than route I.

7.4 Reactions with hydrogen

In 2007, Baiker and coworkers reported a DFT and high pressure spectroscopic investigation on the mechanism of hydrogenation of CO_2 catalysed by a ruthenium dihydride complex.[29] The mechanism proposed is shown in Scheme 30. In the proposed catalytic cycle, *trans*-[Ru(dmpe)$_2$$H_2$] was assumed as the active species. This assumption was based on the experimental results that species that have a dihydrogen bond between a hydride ligand in *trans*-[Ru(dmpe)$_2$$H_2$] and the proton in a formic acid molecule were observed under high H_2 pressure although *cis*-[Ru(dmpe)$_2$$H_2$] is usually expected to be thermodynamically more stable than *trans*-[Ru(dmpe)$_2$$H_2$]. Starting from *trans*-[Ru(dmpe)$_2$$H_2$], CO_2 inserts into a Ru–H bond to form a ruthenium formate *via* a direct hydride abstraction process, which was discussed in Section 2.5 (Scheme 7). Then, an H_2 molecule inserts into the ruthenium–formate bond, leading to the formation of a dihydrogen bonded complex. This insertion



Scheme 29

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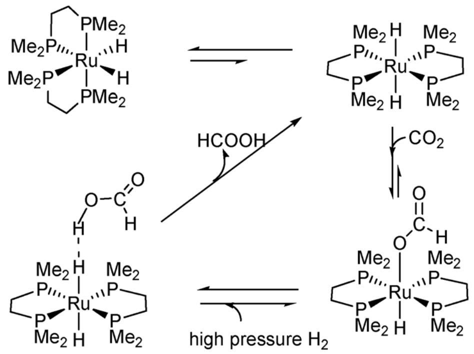
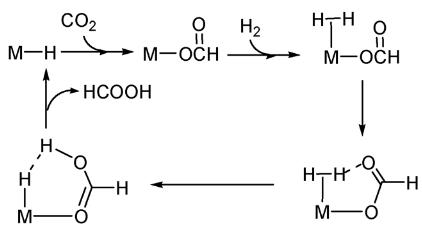


Scheme 30



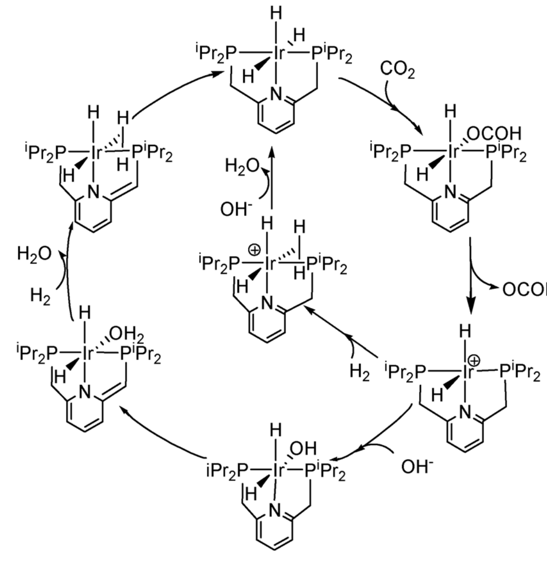
M = $[Ru(PH_3)_3H]$, $[Rh(PH_3)_2]$, $[Rh(PH_3)_2(H_2O)H]^+$

Scheme 31

is the rate-determining step with an activation electronic energy of 25.4 kcal mol^{-1}.

In 2000 and 2002, Sakaki and coworkers reported theoretical studies of Rh(III) and Ru(II)-catalysed hydrogenation of CO_2 to formic acid and compared the reactivity of Rh(III), Ru(II) and Rh(I) complexes.[26,27] The mechanisms are similar irrespective of the metal complexes used (Scheme 31). Metal-hydride species were believed to be the active species. CO_2 inserts into an M–H bond followed by coordination of H_2. Formic acid is formed *via* the six-membered ring σ-bond metathesis. The rate-determining step of hydrogenation of CO_2 catalysed by $Ru(PH_3)_3H_2$ or $Rh(PH_3)_2(H_2O)H_2{}^+$ was found to be the CO_2 insertion [$E_a(Ru(PH_3)_3H_2)$ = 10.3 kcal mol^{-1}, $E_a(Rh(PH_3)_2(H_2O)H_2{}^+)$ = 28.4 kcal mol^{-1}]. For hydrogenation of CO_2 catalysed by $Rh(PH_3)_2H$, isomerization associated with a simple 180° rotation of the –C(O)H moiety along the O–C single bond in the species $Rh(\eta^1\text{-}OC(O)H)(PH_3)_2(H_2)$, which is derived from insertion followed by H_2 coordination, was the rate-limiting step (E_a = 6.1 kcal mol^{-1}). The CO_2 insertion into the 14-electron $Rh(PH_3)_2H$ and the six-membered ring σ-bond metathesis occur with nearly no barrier.

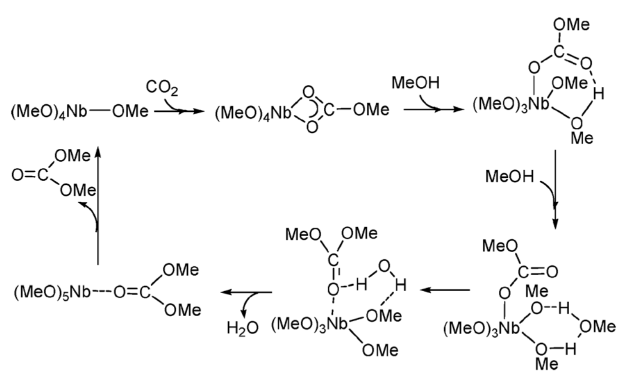
In 2009, Nozaki and coworkers reported that Ir(III)-pincer complexes efficiently catalysed hydrogenation of CO_2.[91] In this experimental study, it was found that the reaction rate is dependent on the H_2 and CO_2 pressure, the strength of the base and the polarity of solvents. The reaction mechanism was also extensively investigated theoretically.[28,30,92] The proposed mechanism on the basis of these theoretical studies can be summarized as in Scheme 32. Starting with a trihydride complex, CO_2 inserts into an Ir–H bond to give an iridium formate *via* a direct hydrogen abstraction process (Section 2.5). Dissociation of the formate ion gives a cationic iridium complex. From the cationic complex, two possible pathways are possible. In the first pathway, coordination of H_2 gives a dihydrogen complex. The dihydrogen complex then reacts with a base to regenerate the trihydride complex. A solvation-corrected activation free energy of 14.4 kcal mol^{-1} was calculated for this rate-determining step. The second pathway involves dearomatization of the pyridine ring of the PNP ligand. Coordination of a hydroxide ion to the cationic iridium complex gives an iridium hydroxide complex. The coordinated hydroxide ion then abstracts a methylene proton, leading to dearomatization of the pyridine ring of the PNP ligand. A solvation-corrected activation free energy of 17.6 kcal mol^{-1} was calculated for this rate-determining step. A ligand exchange of H_2 for H_2O occurs followed by a re-protonation step to re-aromatization of the pyridine ring of the PNP ligand.



Scheme 32

7.5 Reactions with methanol

In 2003, Aresta, Dibenedetto and coworkers found experimentally that $[Nb(OR)_5]_2$ (R = Me, Et, allyl) was capable of catalyzing direct carboxylation of aliphatic alcohols to organic carbonates.[93] In 2006, they investigated its reaction mechanism theoretically and proposed the catalytic cycle shown in Scheme 33. In the theoretical calculations, they employed $Nb(OMe)_5$ as the model catalyst.[51] CO_2 insertion into an Nb–OMe bond was considered as the first step of the catalytic cycle. The methylcarbonate generated from the CO_2 insertion then takes in two molecules of methanol, one directly



Scheme 33

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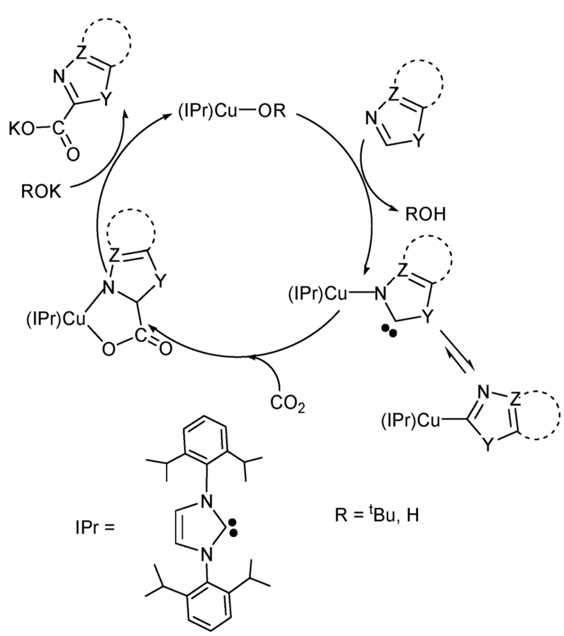
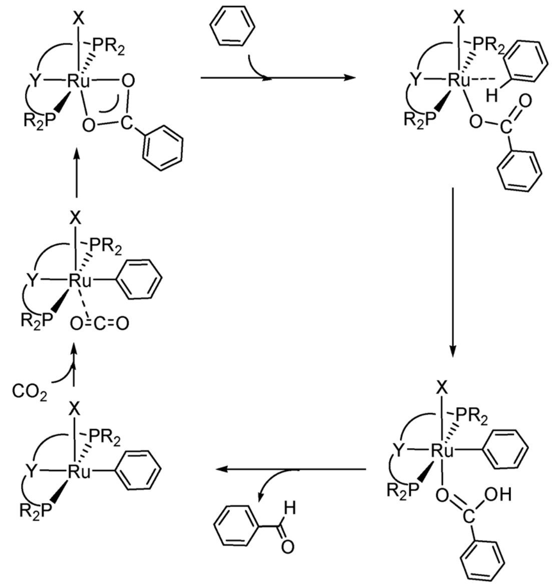
coordinated to the metal center and the other one hydrogen-bonded to the metal-coordinated methanol. From the intermediate that has two methanol molecules, dimethylcarbonate is formed *via* a one-step pericyclic reaction mechanism involving a methyl migration from the hydrogen-bonded methanol to the methylcarbonate ligand. This one-step methyl migration process was calculated to be the rate-determining step with an activation electronic energy of 41.9 kcal mol^{-1}.

7.6 Reactions with arenes

In 2010, Boogaerts and Nolan found experimentally that *N*-heterocyclic carbene gold(I) complexes can catalyze carboxylation of heteroarene C–H bonds.[94] Later, their group, as well as Hou's group, found that *N*-heterocyclic carbene copper(I) complexes can also catalyze the carboxylation of heteroarene C–H bonds.[95] In 2011, Ariafard, Yates and coworkers reported DFT calculations on carboxylation of the heteroarene C–H bonds catalysed by copper(I) complexes.[96] An interesting reaction mechanism was proposed on the basis of the DFT calculations (Scheme 34). Starting with (NHC)Cu–OR as the active species, deprotonation takes place as the first step *via* a four-membered ring transition state with the nitrogen atom in the heteroarene molecule coordinating to the copper metal center and the alkoxy ligand acting as a base to deprotonate the C–H bond of heteroarene. An N-coordinated carbene ligand, derived from the deprotonation, can isomerize to a C-coordinated heteroarylcopper complex. Although the C-coordinated heteroarylcopper complex is more stable than the N-coordinated carbene complex, CO_2 insertion into the C-coordinated heteroarylcopper complex has a much higher activation free energy ($\Delta G^{\ddagger}$ = 30.9 kcal mol^{-1} for benzoxazole) than CO_2 insertion into the N-coordinated carbene complex ($\Delta G^{\ddagger}$ = 20.1 kcal mol^{-1} for benzoxazole with an energy reference to heteroarylcopper complex). This is expected because the N-coordinated carbene ligand acts as a strong nucleophile to attack CO_2 carbon *via* its uncoordinated C end. The CO_2 insertion into the N-coordinated carbene complex can also be viewed



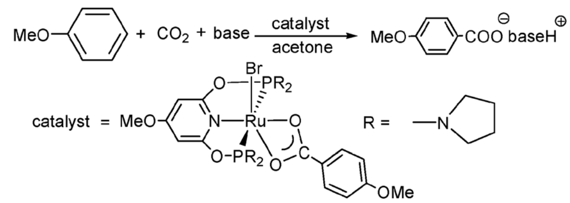
Scheme 34



Scheme 35

as a [2+3] cyclization process. The last step is the release of the carboxylate product by reacting with KOR and regeneration of the (NHC)Cu–OR active species. The rate-determining step is the C–H deprotonation step with an activation free energy of 23.6 kcal mol^{-1} when benzoxazole is used as the substrate.

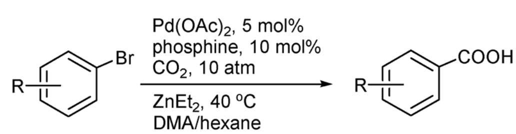
In 2012, Uhe, Hölscher and Leitner carried out DFT studies on carboxylation of arene C–H bonds catalysed by ruthenium pincer complexes.[97] With the aid of DFT calculations and employing the prototypical mechanism (deprotonation followed by insertion) shown in Scheme 35, the authors were able to propose synthetically readily accessible pincer ligands for the catalytic reactions and predicted that the reaction shown in eqn (10) would have the highest turnover frequencies (TOFs).



(10)

8. Perspectives

From what we have discussed above, we see that tremendous progress has been made in understanding reactions of carbon dioxide mediated or catalysed by transition metal complexes. However, it is still too early to say that we have reached a complete picture regarding reactions of carbon dioxide. Here we give a few important experimentally observed examples to illustrate the need of employing further theoretical studies to obtain insights into their mechanisms.



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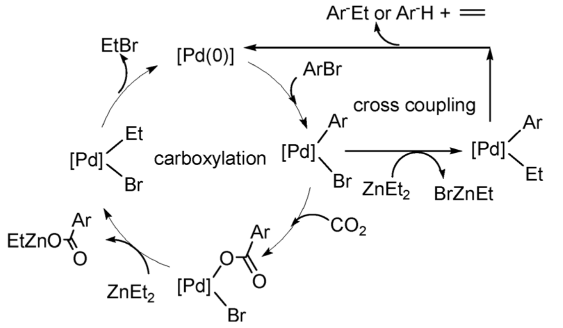
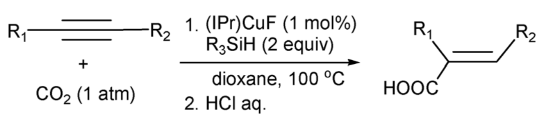


Scheme 36

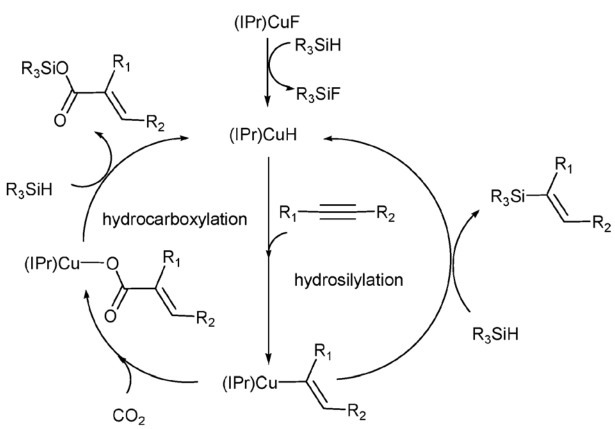
(1) Recently, Martin, Correa and coworkers reported an example of direct carboxylation of readily available aryl bromides with CO_2 *via* Pd catalysis (eqn (11)).[98] It is well known that palladium can catalyze a number of carbon–carbon cross coupling reactions. Therefore, competing catalytic cycles (carboxylation *versus* cross coupling) exist (Scheme 36). The reactions are remarkable in that the corresponding palladium-catalysed Negshi cross coupling reactions did not occur. The important scientific issues are why this reaction is so special and does not give the Negshi cross coupling product and how the CO_2 insertion into the Pd–Ar bond competes favorably against the transmetallation between [Pd](Ar)(Br) and $ZnEt_2$.

(2) In 2011, Tsuji and coworkers reported copper-catalysed hydrocarboxylation of alkynes using CO_2 and hydrosilanes shown in eqn (12).[99] Scheme 37 gives a feasible reaction mechanism for the reactions. Clearly, hydrosilylation competes against the desired hydrocarboxylation reaction. Here, insertion *versus* metathesis is the key scientific issue that needs to be addressed computationally.

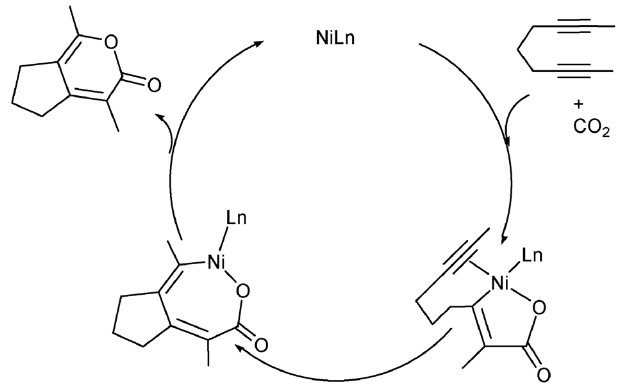


(12)

(3) Copper has also been found to catalyze carboxylative coupling of terminal alkynes, allylic chlorides and CO_2 (eqn (13)).[100] The substrates for the reactions are not limited to allylic chlorides. Propargylic and benzylic chlorides, α-chloro carbonyl compounds also react with terminal alkynes

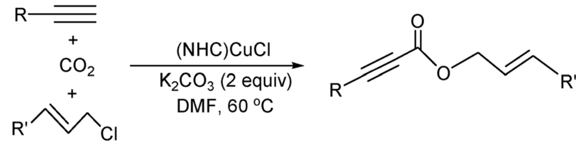


Scheme 37



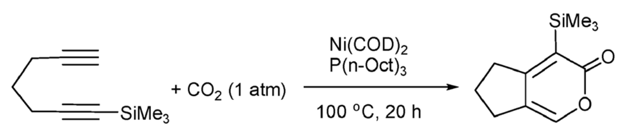
Scheme 38

and CO_2 to give the corresponding carboxylative coupling products. A direct coupling between terminal alkynes and allylic chlorides is possible. Another possible coupling is the reaction between CO_2 and alkynes, a direct C–H carboxylation.[101] All these possible couplings will complicate the reactions and could give undesired products.

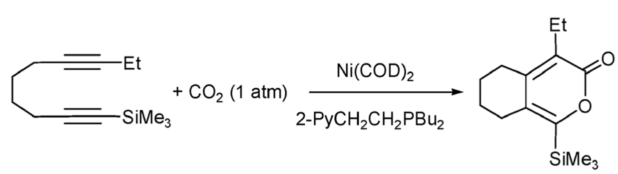


(13)

(4) The interesting regioselectivity of the [2+2+2] reaction of diynes with CO_2 catalysed by $Ni(COD)_2$ in the presence of phosphine ligands (eqn (14) and (15)) was found.[102] For example, in eqn (14), the silyl group in the product is attached to the carbon atom that is bonded to the carbonyl unit. However, in eqn (15), the silyl group in the product does not have the same regioselectivity. Clearly, the reasons for the interesting regioselectivity are not known, although further experiments led to a tentative mechanism shown in Scheme 38.[103] Again, detailed theoretical studies on these reactions are not yet available.



(14)



(15)

9. Conclusions

In this Feature article, we have reviewed theoretical studies of reactions of carbon dioxide. We have focused on (1) insertion reactions into various metal–ligand bonds; (2) oxidative coupling reactions with unsaturated organic substrates; (3) stoichiometric reactions mediated by transition

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metal complexes; and (4) reactions catalysed by transition metal complexes.

The carbon atom in CO_2 is highly electron-deficient, yet it forms strong double bonds with both oxygen atoms. Therefore, it is chemically challenging to activate CO_2 for further chemical transformation. Because of the electron-deficient nature in the carbon center and the high electronegativity of the two oxygen atoms, CO_2 usually acts as an electrophile. Therefore, the reaction partners of CO_2 must be nucleophilic and electron rich.

In CO_2 insertion reactions, metal complexes, with which CO_2 reacts, are usually 16- or 14-electron species in which an available vacant site is available for CO_2 to coordinate with. The CO_2 insertion into a metal–ligand bond of such metal complexes usually occurs *via* a normal insertion mode. Activation free energy barriers for insertion reactions of CO_2 into a metal–ligand bond with a normal insertion mode increase in the order of M–OH (smaller than 10 kcal mol^{-1}) < M–H (*ca.* 10 kcal mol^{-1}) < M–boryl (*ca.* 15 kcal mol^{-1}) < M–hydrocarbyl (10 – 20 kcal mol^{-1}) < M–silyl/germyl/stannyl (*ca.* 30 kcal mol^{-1}).

CO_2 insertion reactions with 18-electron complexes without a ligand pre-dissociation are possible when the insertion occurs into a metal-hydride. Without an initial coordination with a metal center, CO_2 can insert into a metal–hydride bond *via* a direct hydride abstraction mode because of the spherical 1s valence orbital of H. However, free energy barriers *via* a direct hydride abstraction mode are normally high (20–30 kcal mol^{-1}). For CO_2 insertion into a metal–η^1-allyl bond in many 16-electron complexes, an S_E2 activation mode without metal coordination was found to be preferable (over normal CO_2 insertion) *via* interacting directly the π bond of the η^1-allyl ligand with the electron-deficient carbon center of CO_2. Activation free energies with such an S_E2 activation mode were calculated to be around 10 kcal mol^{-1}.

Oxidative coupling reactions mediated by nickel(0) complexes have been studied most extensively. 16-Electron precursor complexes L_2Ni(substrate) are often considered to react with CO_2. The free energy barriers for these oxidative coupling reactions were found to be around 30 kcal mol^{-1}. In oxidative coupling reactions with unsaturated organic substrates, CO_2 acts as an electrophile and unsaturated organic substrates act as nucleophiles. The more electron-rich the unsaturated organic substrates are, the more easily the oxidative coupling reactions take place. The relative electron richness of the two doubly or triply bonded atoms in an unsaturated organic substrate under consideration controls the regioselectivity of oxidative coupling reactions. For example, the terminal carbon of $HC{\equiv}C$–OH is more electron-rich in terms of the π electron density than the internal carbon. As a result, the terminal carbon preferentially couples with the CO_2 carbon. In some special case, steric effect could overturn the expected regio preference. Oxidative coupling reactions with 18-electron Mo or W complexes containing olefin unsaturated substrates as ligands have also been studied. A ligand pre-dissociation is necessary for the complex to coordinate with CO_2 to allow the coupling to occur.

In the section of stoichiometric reactions mediated by transition metal complexes, we have discussed (1) oxygen abstraction reactions, (2) reductive disproportionation reactions, and (3) metal-mediated reactions of CO_2 with other organic substrates. All of these reactions discussed involve C–O bond cleavage of CO_2. One common feature of these reactions is that in most cases the rate-determining step is related to C–O bond cleavage except the reductive cleavage reaction of CO_2 mediated by a bis-β-diketoiminatediron dinitrogen complex where the rate-determining step was found to be the substitution process of CO_2 for N_2. Another common feature of these reactions is that electron-rich metal complexes first take in CO_2 by converting one of the two C=O double bonds to a single bond followed by cleavage.

In the section of reactions of CO_2 catalysed by transition metal complexes, we have discussed (1) reactions with organoboranes, (2) coupling reactions with ethylene, (3) coupling reactions with epoxides, (4) reactions with hydrogen, (5) reactions with methanol, and (6) reactions with heteroarenes.

For reactions with organoboranes, the reaction mechanisms commonly involve CO_2 insertion followed by transmetallation. It is known that boron is highly oxophilic. Therefore, oxygen abstraction leading to CO release was often observed in these reactions. For such reactions accompanying formation of CO, releasing CO occurs from the intermediates derived from the CO_2 insertion and is often the rate-determining step. For coupling reactions with ethylene, the reaction mechanisms normally involve oxidative coupling, β-H elimination, and reductive elimination. For coupling reactions with epoxides, the reaction mechanisms generally involve oxidative addition of epoxide, CO_2 insertion and reductive elimination. In reactions with dihydrogen, dihydrogen molecules are the hydrogen source for formation of metal hydrides from which CO_2 insertion occurs followed by protonation of the formate ligand formed to give the formic acid product. In reactions with methanol, methanol molecules are the source for formation of metal–methoxyl bonds from which CO_2 insertion occurs easily to form organic carbonates. In reactions with heteroarenes, carboxylation of arene C–H bonds occurs normally *via* deprotonation followed by CO_2 insertion.

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